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What can Large Language Models offer to Event-based Vision?

Master Thesis

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Abstract

Vision-Language Models (VLMs) have achieved impressive progress in multimodal reasoning, yet they remain predominantly tailored to conventional frame-based camera data. This dependency limits their applicability in high-speed or low-light environments, where event-based sensing excels. Advancing toward event-aware VLMs has been slowed by two challenges: the lack of large-scale event-text datasets, alongside the fundamental misalignment between sparse, asynchronous event streams and the dense, grid-aligned inputs expected by RGB-trained models.

This work introduces, to our knowledge, the first event-native vision-language model which eliminates frame conversion and leverages event-based pre-training to ground language generation in event sequences. To overcome the data bottleneck, we construct a novel event-text dataset for event stream captioning. Our model explicitly exploits temporal and polarity cues in event streams, which are usually lost in frame-based approaches. By adapting established image-based pre-training strategies to the unique properties of event data, we obtain superior performance on the CIFAR10-DVS benchmark, surpassing previous event-based pre-training methods.

Nomenclature

Notation

W	Sensor width
H	Sensor height
$\boldsymbol{K}$	Key matrix
$\boldsymbol{V}$	Value matrix
$\boldsymbol{Q}$	Query matrix
$\lVert \cdot \rVert$	L^2 norm
$\mathcal{L}$	Loss function
$\mathrm{sim}(\cdot, \cdot)$	Similarity measure

Vectors are written in lower case bold letters ($\mathbf{a}$) and matrices in upper case bold letters ($\mathbf{A}$). Datasets of N samples are denoted as $\{a^{(i)}\}_{i=1}^{N}$, where $a^{(i)}$ is the i-th element of the dataset.

Acronyms and Abbreviations

AI	Artificial Intelligence
DNN	Deep Neural Network
MLP	Multi-Layer Perceptron
LLM	Large Language Model
VLM	Vision-Language Model
SSL	Self-Supervised Learning
VQA	Visual Question Answering
NLP	Natural Language Processing

Chapter 1

Introduction

Event cameras are bio-inspired sensors [7, 44, 63] that operate fundamentally differently from traditional frame-based cameras. Rather than capturing full-frame luminance at fixed intervals, they trigger asynchronously in response to relative brightness changes at each pixel. The output is a sparse stream of events, each encoding the time, location, and polarity (sign) of the brightness change. Enabled by the event-driven sensing mechanism, event cameras achieve microsecond-level temporal resolution, high dynamic range, low power consumption, and strong resistance to motion blur — effectively addressing main limitations of conventional cameras.

Leveraging these advantages, event-based methods have demonstrated promising performance in tasks, such as object recognition [36, 62], detection [28, 40, 54], tracking [22, 24], and optical flow estimation [3, 26, 80]. However, despite the rich spatiotemporal cues inherent in event streams, their integration into high-level semantic tasks — particularly those involving language grounding and open-world understanding — remains underexplored.

Vision-Language Models [11, 41, 48, 55, 72] have gained significant attention in artificial intelligence research, building on the success of Large Language Models [8, 17, 56, 64]. VLMs are designed to process and understand both visual (images or videos) and textual data. Unlike traditional models that focus on a single modality, Vision-Language Models combine information from both, allowing them to interpret, generate, and reason across vision and language. This capability mirrors the way humans naturally process information from multiple sources simultaneously.

Modern VLMs have exhibited remarkable advancements in scene understanding, complex reasoning, and visual question answering [1, 46]. These achievements are primarily driven by large-scale image-text datasets [45, 75, 81] and architectures tailored to conventional frame-based cameras. Such dependence limits the model performance in high-speed or low-light environments where event-based sensing excels. Despite the aforementioned advantages of the latter, the development of event-aware VLMs [43, 47, 73] remains surprisingly limited.

This research gap stems from two key challenges:

1. **The dataset bottleneck.** Unlike the abundant image-text resources available for RGB-based VLMs, the event vision community lacks comparable large-scale event-text datasets, limiting supervised training. Existing event datasets [36, 39, 50] focus primarily on classification tasks rather than rich vision-language alignment.

2. **The representation mismatch.** Event data is sparse, asynchronous, and polarity-sensitive, while VLMs expect dense, grid-aligned RGB inputs. Current approaches [43, 47, 73] attempt to bridge this gap by converting events into frame-like synchronous representations, that inevitably discard crucial temporal and polarity information contained in events.

This work introduces, to our knowledge, the first **event-native vision-language model**, which eliminates frame conversion and leverages event-based pre-training to ground language generation in event data. The model is finetuned on a novel event-text dataset for event stream captioning. The dataset is constructed by simulating event streams from an existing video-caption corpus to create paired event data and textual descriptions for cross-modal alignment. Unlike frame-based pipelines that are prone to image biases while omitting essential polarity and temporal cues, our model makes explicit use of this information and is trained entirely within the event domain. By adapting image-based pre-training strategies to the unique characteristics of event data, our approach outperforms previous event-based pre-training methods on the CIFAR10-DVS benchmark.

1.1 Related Work

Our study centers on decisions regarding event representation, encoder backbone, pre-training procedure, and text decoder, which together determine how a vision-language model is formed. We therefore survey prior research on event-based representations and associated architectures, established image-based pre-training and VLM techniques, as well as their extensions to event-based vision, in order to contextualize our approach.

1.1.1 Event Representations

Event cameras asynchronously record per-pixel brightness changes and output a stream of events $\mathcal{E} = \{e_i = (x_i, y_i, t_i, p_i)\}$. Each event e_i encodes the spatial coordinates (x_i, y_i) of the activated pixel, its timestamp t_i, and polarity $p_i \in \{-1, 1\}$, indicating relative decrease or increase in brightness. While individual events carry minimal information due to the lack of global context, they are often aggregated into structured representations to extract meaningful features for downstream tasks [21].

Individual events $e_i = (x_i, y_i, t_i, p_i)$ are used as an input to event-by-event processing methods, such as probabilistic filters [23] and Spiking Neural Networks [51]. Importantly, each event is interpreted not in isolation, but in the context of a dynamically evolving state [21]. This state may be built from the history of previously observed events or informed by prior knowledge about the scene, task, or sensor dynamics. As a result, the response to a new event can

depend heavily on what the system has already observed or inferred, enabling temporal continuity and predictive reasoning.

Frame-based representations. A straightforward approach to adapting events for existing deep learning algorithms is to convert asynchronous event streams into synchronous 2D image representations, which can then serve as inputs to Deep Neural Networks. Based on the methodology used to construct the frames, prevailing techniques can be categorized into three main types: (i) polarity-based stacking [49], (ii) timestamp-based stacking [66], (iii) event-count-based stacking [80]. Despite the statistical differences between event frames and natural images, numerous studies [14, 57] have shown that such input adaptation can yield strong performance. However, these representations remain suboptimal for event-based paradigm, as they sacrifice sparsity, risk losing temporal or polarity information, and exhibit high sensitivity to the number of events per frame.

Voxel grid is a 3D histogram of events, where each voxel corresponds to a specific spatial location (pixel) and temporal interval, effectively creating a volumetric encoding of event occurrences across both space and time. Unlike frame-based methods, this representation avoids collapsing the temporal information on a 2D grid [15, 16]. However, such volumetric encoding implies cubic growth in memory requirements with increasing spatial and temporal resolution, making high-resolution voxel grids computationally demanding. Furthermore, due to the sparse and asynchronous nature of the event data, most voxels remain empty, leading to inefficient memory usage and processing overhead.

Token-based representations have emerged as a central paradigm in modern deep learning, particularly for processing structured and sequential data. In Natural Language Processing, models operate on tokens, typically words or subwords, that serve as discrete, context-aware information units. This concept was subsequently extended to computer vision by [18], who showed that the standard transformer could be seamlessly applied to image data with minimal modifications by dividing images into fixed-size patches, projecting each patch into a linear embedding, and treating them as visual tokens.

In event-based vision, various tokenization strategies have been proposed. The most common approach, frame-based tokenization [59, 60, 70] first converts event streams into frames before splitting them into patches. Although computationally efficient, this technique inherits the fundamental drawbacks of the frame-based representations mentioned above. In contrast, event-level tokenization [35] treats each individual event as a token, embedding its spatiotemporal attributes into a fixed-dimensional vector. This preserves the high temporal precision of event data, but may face scalability challenges for dense event streams.

A third strategy, group tokenization [53] explicitly clusters events by their timestamp and polarity, standardizing variable-length event streams into fixed-size tokens while preserving temporal and spatial structure. The method's advantages are twofold: it avoids the temporal quantization loss of frame-based methods while maintaining computational efficiency compared to per-event processing. Rigorous ablation studies validate these benefits, showing consistent accuracy gains across tasks — clear evidence that proper handling of event-specific characteristics is crucial for performance.

1.1.2 Transformers for Event-based Vision

The self-attention mechanism, introduced in [65], has become a cornerstone of modern DNNs. Following its revolutionary impact on NLP, this mechanism found new utility in computer vision with the Vision Transformer (ViT) [18], demonstrating versatility of attention-based models for multimodal learning.

Motivated by the widespread adoption of transformers, a growing body of research explores their suitability for event-based vision. Recent studies have deployed these models across diverse tasks including classification [59, 60], object detection [27], and video reconstruction [71], with further works focusing on direct feature extraction from event representations [27, 70, 76]. Yet, many transformer-based approaches overlook critical event characteristics such as time and polarity, leaving their potential untapped.

Group Event Transformer (GET) [53] addresses these limitations through several architectural innovations. First, the Event Dual Self-Attention (EDSA) block separately processes spatial and temporal-polarity features extracted from Group Tokens (Section 1.1.1) through dedicated attention mechanisms, enabling their effective communication. Second, the Group Token Aggregation (GTA) module employs overlapping group convolutions that progressively merge tokens to integrate and decouple information in both domains. As a result, GET establishes new state-of-the-art performance and surpasses existing methods on multiple classification [2, 34, 39, 62] and detection [12, 54] benchmarks.

1.1.3 Vision-Language Models

Current training methodologies for VLMs primarily fall into three categories [6]: (i) contrastive learning, (ii) masked modeling, and (iii) generative approaches. Each paradigm offers distinct advantages in aligning visual and textual representations while addressing different downstream tasks.

Contrastive training is a commonly utilized strategy that relies on positive and negative example pairs to learn meaningful representations. The model is trained to maximize similarity between matching image-text pairs (positives) while minimizing similarity between mismatched ones (negatives). A prominent work is Contrastive Language-Image Pre-training (CLIP) [55], which employs the InfoNCE loss to align visual and textual embeddings in a shared latent space. CLIP defines positive pairs as an image and its corresponding caption, while negative pairs consist of the same image paired with unrelated captions from the same batch. By training on 400 million web-collected multimodal samples, CLIP showed remarkable zero-shot classification transfer capabilities.

Masked modeling is another effective training strategy, where the model learns by reconstructing masked portions of input data. For VLMs this can involve reconstructing masked image patches given an unmasked text caption or predicting masked words in a caption conditioned on an unmasked image. For instance, FLAVA [61] combines multimodal and unimodal masked modeling objectives along with a contrastive loss. Critically, such joint optimization forces the model to develop aligned representations that preserve both modality-specific semantics and cross-modal relationships. As a consequence, FLAVA

excels across 35 diverse vision, language, and multimodal benchmarks, demonstrating robust understanding across domains.

Generative VLMs, unlike contrastive or masked models that focus solely on representation learning, synthesize text directly from images. However, their autoregressive nature and extensive training requirements make them computationally expensive, while the scarcity of open-source implementations further limits accessibility. A notable exception is DeepSeek-VL [48], which combines large-scale pre-training with an efficient three-stage pipeline (adaptor warmup, joint vision-language pre-training, and supervised finetuning) to achieve state-of-the-art or competitive results across a wide range of vision-language benchmarks. By open-sourcing their models, DeepSeek bridges the gap between proprietary and community-driven innovation in multimodal AI.

Another influential work is BLIP [42], which unifies contrastive, matching, and generative objectives within a single framework. BLIP introduces a bootstrapped captioning-and-filtering strategy (CapFilt) that leverages synthetic captions and noise-robust filtering to improve training data quality. This combination enables BLIP to learn versatile multimodal representations and achieve strong performance across captioning, retrieval, and visual question answering, while also supporting both understanding and generation within the same architecture.

1.1.4 Event-based Vision-Language Models

While event understanding remains in its early stages compared to its well-established image-based counterpart, recent work demonstrates promising progress. For instance, EventCLIP [73] transfers the pre-trained CLIP [55] model to enable zero-shot and few-shot event-based object recognition. In contrast, EventBind [79] goes further by jointly aligning images, text, and events in a shared representation space. It introduces a dedicated event encoder to leverage data-specific properties while retaining CLIP as an image encoder, facilitating few-shot recognition and cross-modal event retrieval. These methods have proved effective, but face two critical shortcomings: (i) inability to leverage the world knowledge embedded in LLMs, and (ii) restriction to discriminative tasks. Consequently, they cannot handle complex scenarios requiring either sophisticated event interpretation (e.g., contextual reasoning) or generative capabilities (e.g., event-conditioned dialogue).

EventVL [43] and EventGPT [47] mark significant advances in event-aware generative vision-language models. EventVL employs a compact 2.3B-parameter design to offer fine-grained semantic understanding through event captioning and dialogue tasks. Its innovative use of hierarchical spatiotemporal splits and dynamic alignment successfully bridges event streams with image-text representations, excelling in zero- and few-shot captioning while retaining compatibility with resource-constrained hardware.

Conversely, EventGPT adopts a distinct approach by eliminating a separate image encoder and instead focusing on a three-stage training paradigm including vision-language alignment, event-language alignment, and instruction finetuning similar to Deepseek-VL. This framework gradually fuses event data with LLMs to support reasoning, VQA, and downstream tasks via task-specific heads. With

larger 7B or 13B models, EventGPT shows stronger results in challenging environments (e.g., tunnel hazard detection) and enables versatile applications via integrated task-specific modules for object detection and instance segmentation.

It is important to emphasize that all aforementioned models, despite their innovations, remain fundamentally constrained by their reliance on frame-based event representations. Furthermore, neither EventVL nor EventGPT implementation was publicly available at the time of study, preventing direct validation or extension of their work in our research.

1.1.5 Pre-training

Pre-training is a foundational phase in transformer-based models, enabling them to learn broad, transferable representations from large-scale unlabeled data. During this stage, models acquire semantic, syntactic, and structural knowledge that enhances their ability to generalize to unseen tasks. The process typically consists of two main steps: self-supervised pre-training on extensive corpora, followed by finetuning for downstream applications. Unlike traditional supervised learning, which relies on costly and time-consuming manual annotation, self-supervised methods design proxy objectives that derive supervision from the data itself. This makes pre-training especially beneficial in conditions where labeled data is scarce or expensive to obtain. By leveraging pre-trained representations, models adapt to new tasks with minimal task-specific data while staying robust to domain shifts.

This principle manifests differently across modalities. In NLP, two dominant paradigms have emerged: (i) masked language modeling, as shown in BERT [17], which learns bidirectional representations by predicting randomly masked tokens within their full context, and (ii) autoregressive prediction, implemented in the GPT series [8, 56], where strictly causal attention masks enforce left-to-right token prediction, enabling generative capabilities. In computer vision, while supervised pre-training of ViTs [18] on extensive datasets like ImageNet-21k or JFT-300M initially set performance benchmarks, self-supervised approaches like DINO [9] have demonstrated the remarkable effectiveness of learning from unlabeled data. Through ViT-based teacher-student knowledge distillation across augmented image views, DINO not only matches supervised performance on downstream tasks but spontaneously develops object-level segmentation capabilities without explicit training.

Another influential contribution to self-supervised learning for computer vision is SimCLR [10], a contrastive framework that maximizes agreement between augmented views of the same image while pushing apart representations of different images in the latent space. Its strength lies in its simplicity, requiring only (i) stochastic data augmentations, (ii) a projection head to map representations to a contrastive space, and (iii) a contrastive loss (NT-Xent) to measure similarity between positive pairs. Notably, the authors emphasize the central role of carefully composed augmentations, non-linear projections, and large batch sizes. Without the need for memory banks or custom architectures, SimCLR significantly outperformed previous methods, establishing a new state of the art.

1.1.6 Event-based Pre-training

Self-supervised learning for event cameras represents a highly promising research direction, addressing the critical challenge of scarce labeled event data by leveraging the intrinsic properties of event streams. Unlike traditional RGB-based SSL, event-based SSL must account for the inherent characteristics of event sequences such as high temporal resolution, sparsity, and asynchronous nature, necessitating novel methodologies tailored to this modality.

To our knowledge, [74] presents the first comprehensive exploration of event-based SSL. The work develops specialized event augmentations combined with a contrastive learning pipeline, that simultaneously enforces alignment between embeddings of matching event images, as well as between paired event and RGB images. This dual-alignment strategy bridges the gap between event-based and conventional visual representations. On the N-ImageNet [36] benchmark, the proposed technique improves upon image-domain transfer, underscoring the advantages of event-based pre-training.

Subsequent works have studied predictive masking objectives to learn robust latent features from event data. For instance, [37] adapts BERT-style pre-training to event histograms by masking and reconstructing local patches, while [33] proposes a disentangled framework for voxel-based event representations, reconstructing the masked voxels given the visible ones. Unlike the aforementioned techniques, the latter method eliminates the need for paired image-event data and avoids frame-based representations, thereby improving upon its predecessors. Together, these approaches reflect the growing interest in and benefits of adapting self-supervised learning to event-based vision.

Chapter 2

Methodology

As discussed in Section 1.1, the rise of large language models and subsequently visual language models has been largely driven by the success of attention mechanisms and transformer architectures [65]. This progress has recently extended to event-based vision, with works like EventVL [43], EventGPT [47], and EventCLIP [73], which mark the first steps toward VLMs tailored for event data. However, these approaches rely on image pre-trained backbones to transfer knowledge from the image domain, thereby compensating for limited event-specific resources. While this strategy is pragmatic given the availability and abundance of pre-trained image-based models and the demonstrated effectiveness of frame-based event representations, it fundamentally disregards the core characteristics of event streams: asynchronicity, temporal precision, and per-pixel polarity information.

Event data is inherently self-sufficient, yet current methods fail to exploit its native properties. The optimal solution would involve a VLM that operates entirely within the event domain and is completely decoupled from the image modality, free from frame-based representations and trained end-to-end exclusively on event data.

This chapter introduces our approach to developing an event-native VLM that operates on event streams without relying on image-based pre-training or frame-based conversions. We begin by presenting the proposed architecture, designed to fully exploit the asynchronous and sparse nature of event data. Next, we describe the construction of a dedicated event-language dataset, carefully curated to support research and drive progress in this field. Finally, we provide a comprehensive overview of the training pipeline, which involves a two-stage encoder pre-training process followed by cross-modal finetuning.

2.1 Architecture

We propose a framework consisting of three building blocks: (i) an **event encoder** that extracts features from event streams for downstream processing, (ii) a **cross-modal alignment module** that maps these features into a language-compatible space, and (iii) a **pre-trained text decoder** that generates text

conditioned on the aligned representations. This section describes each unit in turn, along with the rationale behind its selection.

2.1.1 Event Encoder

Unlike conventional methods that force event data into image-compatible architectures, our model is built upon a transformer encoder based on core ideas of event-based vision. Group Event Transformer (GET) [53] fulfills this need through the event-specific design: its token representation explicitly encodes temporal information and event polarity, while the novel Event Dual Self-Attention (EDSA) mechanism is purpose-built to handle the resulting spatiotemporal embeddings effectively. GET's combination of architectural innovations yields state-of-the-art accuracy in classification and detection benchmarks despite operating with just 4.5 million parameters in a 3-stage network. Given these advantages, GET emerges as a compelling choice for our event encoder. The following paragraphs provide a detailed description of its group token representation and EDSA layer.

Group Token. Event streams recorded by an event camera with resolution $W \times H$ are stored as a tuple $(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{t}, \boldsymbol{p})$, where $(\boldsymbol{x}, \boldsymbol{y})$ denote the pixel coordinates, $\boldsymbol{t}$ the timestamps, and $\boldsymbol{p}$ the polarities of individual events. To enable structured processing, the temporal dimension is discretized into K intervals, and the spatial domain is divided into non-overlapping patches of size $P \times P$. With $t_{\min} = \min(\boldsymbol{t})$ and $t_{\max} = \max(\boldsymbol{t})$, each event is subsequently encoded using three components: the discretized timestamp $\boldsymbol{d}_t$, the intra-patch rank $\boldsymbol{r}$, and the patch-level position $\boldsymbol{s}$, expressed as

$$\begin{cases} \boldsymbol{d}_t = \left\lfloor K \cdot \dfrac{\boldsymbol{t} - t_{\min}}{t_{\max} - t_{\min} + 1} \right\rfloor, \\ \boldsymbol{r} = \lfloor \boldsymbol{x} \bmod P \rfloor + \lfloor \boldsymbol{y} \bmod P \rfloor \cdot P, \\ \boldsymbol{s} = \left\lfloor \dfrac{\boldsymbol{x}}{P} \right\rfloor + \left\lfloor \dfrac{\boldsymbol{y}}{P} \right\rfloor \cdot \dfrac{W}{P}, \end{cases} \tag{2.1}$$

where $+1$ ensures numerical stability and avoids division by zero.

Then, the polarity along with the three arrays above are combined into a single 1D vector, formulated as

$$\boldsymbol{l} = (K \cdot H \cdot W) \cdot \boldsymbol{p} + (H \cdot W) \cdot \boldsymbol{d}_t + \left(\frac{H \cdot W}{P^2} \right) \cdot \boldsymbol{r} + \boldsymbol{s}. \tag{2.2}$$

Next, two 1D vectors of length $H \cdot W \cdot 2K$ are constructed via a bin count operation. The first one uses weights of $\mathbf{1}$ and the second utilizes normalized relative timestamps computed as $(\boldsymbol{t} - t_{\min})/(t_{\max} - t_{\min})$. These arrays are concatenated and reshaped into a structured output with dimensions $\left(\frac{H}{P} \cdot \frac{W}{P}\right) \times \left(2K \cdot 2P^2\right)$. A 3×3 group convolution, followed by an MLP, hierarchically refines this representation maintaining spatiotemporal relationships.

The resulting Group Tokens have dimensions $\left(\frac{H}{P} \cdot \frac{W}{P}\right) \times (G \cdot C)$, where G is the number of groups, each corresponding to distinct combinations of time interval and polarity, while C indicates the channel number of each group.

Event Dual Self-Attention Layer. In the EDSA layer, the initial step involves partitioning the feature maps into M non-overlapping windows, each of size S, to compute parallel self-attention. Each window of shape $S \times (G \cdot C)$ undergoes Spatial Self-Attention (SSA) to capture spatial context. The SSA operation is formally defined as

$$\begin{aligned} \boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V} &= \boldsymbol{X}\boldsymbol{W}^Q, \boldsymbol{X}\boldsymbol{W}^K, \boldsymbol{X}\boldsymbol{W}^V \\ \mathrm{SSA}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) &= \mathrm{Softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^T}{\sqrt{G \cdot C}} + \boldsymbol{B}_p\right)\boldsymbol{V}, \end{aligned} \tag{2.3}$$

where $\boldsymbol{W}^Q, \boldsymbol{W}^K, \boldsymbol{W}^V \in \mathbb{R}^{(G \cdot C) \times (G \cdot C)}$ are learned linear projection matrices to transform input into query $\boldsymbol{Q}$, key $\boldsymbol{K}$, and value $\boldsymbol{V}$ along the channel dimension. The term $\boldsymbol{B}_p$ denotes the relative position bias (RPB).

The output feature maps from SSA are transposed to a shape $(G \cdot C) \times S$ to enable application of Group Self-Attention (GSA). This mechanism is designed to model temporal and polarity-aware contextual dependencies among the Group Tokens, thereby enhancing the semantic representation. The GSA operation is described as follows

$$\begin{aligned} \boldsymbol{Q}_g, \boldsymbol{K}_g, \boldsymbol{V}_g &= \boldsymbol{X}^T\boldsymbol{W}_g^Q, \boldsymbol{X}^T\boldsymbol{W}_g^K, \boldsymbol{X}^T\boldsymbol{W}_g^V \\ \mathrm{GSA}(\boldsymbol{Q}_g, \boldsymbol{K}_g, \boldsymbol{V}_g) &= \mathrm{Softmax}\left(\frac{\boldsymbol{Q}_g\boldsymbol{K}_g^T}{\sqrt{S}} + \boldsymbol{B}_g\right)\boldsymbol{V}_g, \end{aligned} \tag{2.4}$$

where $\boldsymbol{W}_g^Q, \boldsymbol{W}_g^K, \boldsymbol{W}_g^V \in R^{S \times S}$ are learned linear projection matrices to transform input along the spatial dimension. The term $\boldsymbol{B}_g \in \mathbb{R}^{(G \cdot C) \times S}$ denotes relative group bias (RGB).

The structure of the EDSA layer is illustrated in Figure 2.1.

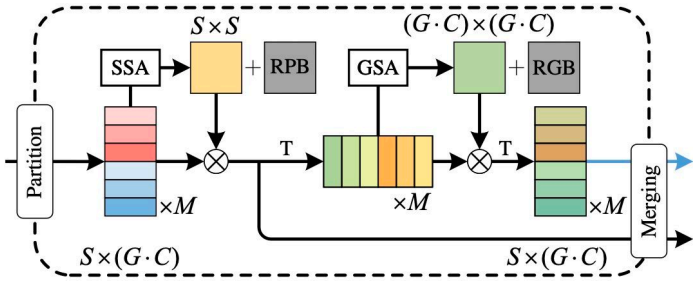

Figure 2.1: Event Dual Self-Attention Layer. Source: [53].

As a result, it produces two feature maps: one obtained from SSA alone (black arrow), and another from joint utilization of SSA and GSA (blue arrow). The EDSA layer ensures effective feature communication in both the spatial and temporal-polarity domains.

Figure 2.2 shows the overview of a 3-stage GET network initialized with a group number of 6. The grouped tokens are progressively fused into larger groups, allowing the features to gradually acquire larger spatial and temporal-polarity receptive fields, which improves object characterization. Thus, GET fully exploits crucial event attributes alongside preserved information decoupling.

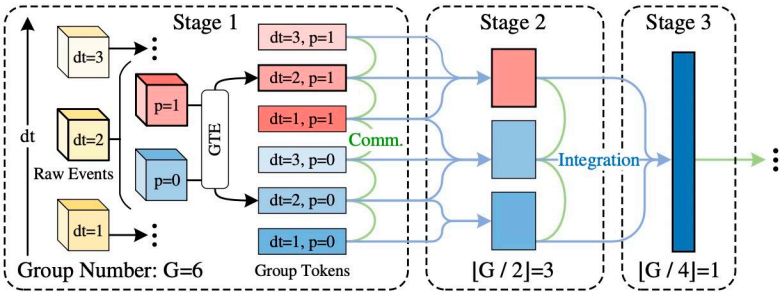

Figure 2.2: 3-stage GET network with a group number of 6. Source: [53].

2.1.2 Cross-Modal Alignment Module

The cross-modal alignment module in a VLM functions as a bridge between visual and textual information by projecting both modalities into a unified representational space. There, semantic relations can be directly compared, which promotes coherence between the two domains. Consequently, elements rooted in a common concept, for example, the textual token "dog" and the event stream region corresponding to a dog, are located in close proximity within this joint latent space. In our proposed framework, the alignment module receives the output features generated by the event encoder and processes them through a projection network consisting of linear transformations, non-linear activation function GELU (Gaussian Error Linear Unit) [30], and normalization layers. The network maps the features to a 512-dimensional vector and then projects them to 768, thereby maintaining consistency with the embedding size required by the text decoder.

2.1.3 Pre-trained Text Decoder

We initially explored larger language models, including DeepSeek LLM 7B [4] and TinyLlama-1.1B [77], chosen for their accessibility and modest parameter size compared to frontier-scale LLMs. However, our experiments uncovered a capacity imbalance: the LLMs proved disproportionately strong relative to the vision encoder. Consequently, they produced fluent yet hallucinatory text that effectively minimized the training loss while neglecting the visual input.

Thus, we selected a smaller, 162 million parameter BLIP [42] text decoder to better match the capacity of the event encoder and ensure tighter visual grounding. Designed for efficiency, it replaces bidirectional self-attention with causal self-attention and conditions on visual embeddings via cross-attention. This setup preserves the autoregressive nature of text generation while enabling dynamic integration of event features. Compared to larger LLMs, the decoder offers a better balance of model capacity and modality alignment, making it well-suited for lightweight vision-language systems. The architecture and objectives of BLIP appear in Figure 2.3. "ITC", "ITM", and "LM" denote the Image-Text Contrastive, Image-Text Matching, and Language Modeling losses, respectively.

Here, cross-attention provides a clear advantage over early fusion strategies that concatenate event and text tokens before feeding them to a language model.

By explicitly attending to visual embeddings at each decoding step, the model preserves the structure of language modeling while integrating information from the encoder. This approach allows for more precise alignment, prevents modality imbalance, and avoids dilution of the signal that often occurs with token fusion.

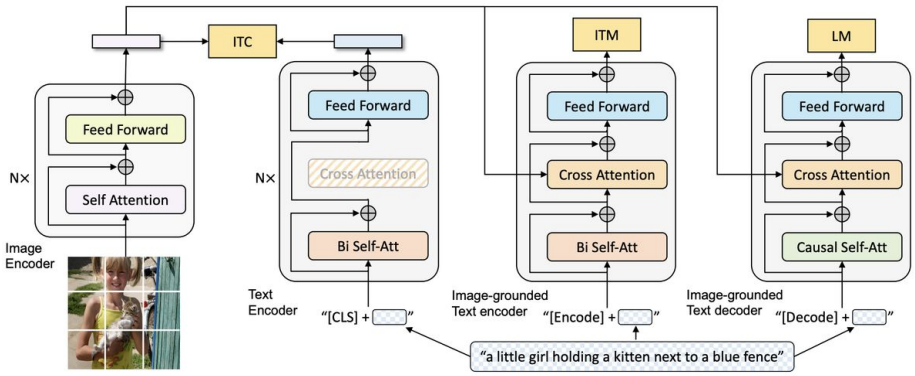

Figure 2.3: BLIP architecture and objectives. Source: [42].

2.2 Event-Text Dataset

Event data remains inherently scarce due to a combination of factors. Firstly, **interpretability**: although event cameras are designed to mimic the way the human eye captures visual information, their raw output is not directly interpretable by human vision unless converted into a specialized representation. Consequently, event cameras are primarily employed as sensors rather than as imaging devices. Secondly, **accessibility**: while event cameras have become more affordable in recent years, their accessibility still falls short compared to frame-based counterparts. This directly leads to a third factor, **constrained adoption**, which in turn hinders the large-scale data collection necessary to overcome scarcity.

Most publicly available event-based vision datasets containing textual annotations are restricted to classification tasks, predominantly consisting of neuromorphic adaptations of established image classification benchmarks. These datasets were typically created by mounting an event camera on a motorized setup and automatically moving it while recording images displayed on a monitor [34, 36, 50]. Because the original labels from the source datasets remained applicable, little manual annotation was necessary.

This reflects a broader issue: even if data acquisition challenges are addressed, annotating event streams remains difficult — particularly when detailed descriptions of object attributes, dynamic actions, or environmental conditions are required. Large-scale image-text datasets are commonly built either by scraping co-occurring image-text pairs from the web or through manual annotation via crowdsourcing platforms. However, these methods are not feasible for event data: it cannot be passively scraped in the same manner, and manual labeling first demands converting the event sequences into a human-interpretable format, adding considerable complexity.

The scarcity of event-text datasets poses a major bottleneck for developing scalable vision-language models capable of comprehensive event-based scene understanding. Recent works EventVL [43] and EventGPT [47] have attempted to address this limitation by introducing new datasets. EventVL leverages the InternVL2-76B [11] foundation model to generate textual descriptions for samples from DSEC [25], N-ImageNet [36], N-Caltech101 [50], and HARDVS [68], exploiting image-event and video-event correspondences to create a dataset of 1.4 million samples. EventGPT, on the other hand, contributes the N-ImageNet-Chat dataset, consisting of 1 million samples annotated via a human-assisted, semi-automated pipeline. At the time of our study, neither dataset was publicly available, which prevented their use in our research.

We construct an event-text dataset derived from InternVid [69], a large-scale video-centric multimodal dataset designed to facilitate the learning of robust and transferable video-text representations for multimodal understanding and generation. InternVid comprises over 7 million public YouTube videos with an average duration of 6.4 minutes, spanning 16 distinct topics. In total, it encompasses nearly 760,000 hours of footage, segmented into 234 million video clips, each paired with detailed textual descriptions amounting to 4.1 billion words. The majority of these clips are short in duration, with 85% falling within the 0-10 second range.

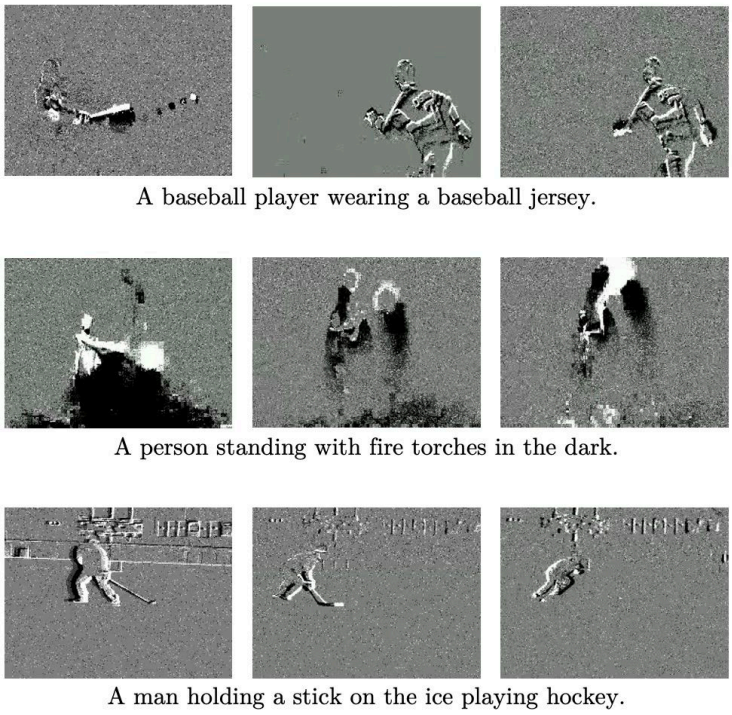

Figure 2.4: Examples from the neuromorphic InternVid dataset.

For meaningful event simulation, short video clips are crucial because they likely capture a single dynamic object or scene. Since event cameras detect brightness changes, input videos must contain consistent motion without scene cuts

or multiple camera angles, common in public videos but disruptive for event simulation. If the video includes abrupt scene changes, the simulated events become noisy and lose their physical meaning.

Due to licensing constraints, the original dataset does not distribute video files directly, but rather provides structured metadata consisting of YouTube video IDs, timestamps, and associated captions. Thus, our first step was to retrieve the actual video clips by scraping YouTube using `yt-dlp` library, while respecting the platform's Terms of Service as well as the per-video license restrictions. We set a 10-second maximum video duration to achieve realistic event simulations within practical memory limits. After collecting 104,000 video clips, we generated corresponding event sequences using the v2e [32] simulator. Quality control was subsequently performed by identifying and removing empty or abnormally short event streams (i.e., fewer than 1,000 events) resulting from corrupted simulations or static videos. The cleaning step yielded a final dataset of 102,000 event-text pairs. Each pair combines an event stream with its associated caption from InternVid, with selected examples displayed in Figure 2.4. Throughout this work, we refer to the constructed dataset as "neuromorphic InternVid".

2.3 Training Stages

Unimodal Contrastive Pre-training. Pre-training has become an established practice in modern VLMs, enabling them to learn generalizable representations from large-scale data before task-specific adaptation. For event-based vision, where labeled data is inherently scarce, self-supervised pre-training is not just beneficial but necessary. Yet existing event-aware VLMs sidestep this principle: they repurpose image pre-trained backbones and skip event-specific self-supervised learning entirely, proceeding directly to event-text alignment. This oversight constrains event encoders to rely on priors mismatched with the sparse, asynchronous nature of event data. Our work enforces a disciplined pre-training regime: we train the event encoder from scratch via self-supervision, compelling it to discover native spatiotemporal and semantic structure in the event streams before any cross-modal finetuning.

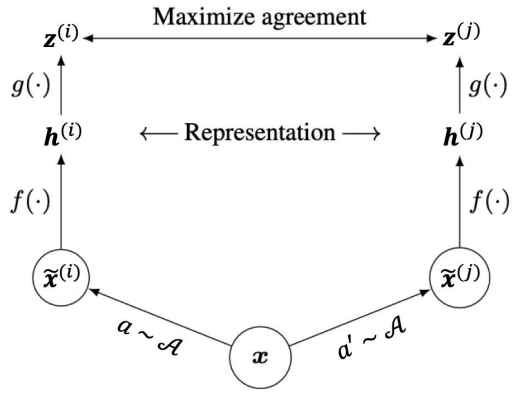

Figure 2.5: SimCLR framework. Source: [10].

We implement a self-supervised objective inspired by SimCLR: A Simple Framework for Contrastive Learning of Visual Representations [10], repurposing it for event-based vision. As depicted in Figure 2.5, two augmentation operators $a \sim \mathcal{A}$ and $a' \sim \mathcal{A}$ are independently sampled from the same family of transformations and applied to the input to produce a pair of correlated views. A shared encoder $f(\cdot)$ and a projection head $g(\cdot)$ are jointly trained to maximize agreement between these views using a contrastive loss. Once training is complete, the projection head is discarded, and the encoder $f(\cdot)$ is used to extract embeddings $\boldsymbol{h}$ for downstream tasks.

To train the encoder, we use the normalized temperature-scaled cross entropy loss, known as NT-Xent. It promotes representations of augmented views derived from the same input to be close in latent space, while repelling those from different inputs within a batch. In a mini-batch of size N, each sample is augmented twice, yielding $2N$ total examples. For a positive pair of embeddings $(\boldsymbol{z}^{(i)}, \boldsymbol{z}^{(j)})$, where $\boldsymbol{z}^{(i)} = g\big(f\big(a(\boldsymbol{x})\big)\big)$ and $\boldsymbol{z}^{(j)} = g\big(f\big(a'(\boldsymbol{x})\big)\big)$, the remaining $2(N-1)$ samples serve as negatives. The cosine similarity between feature vectors $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$ is defined as $\mathrm{sim}(\boldsymbol{u}, \boldsymbol{v}) = \boldsymbol{u}^\top \boldsymbol{v} / ||\boldsymbol{u}||\,||\boldsymbol{v}||$. Then the loss function for (i, j) is given by

$$\ell(i,j) = -\log \frac{\exp(\mathrm{sim}(\boldsymbol{z}^{(i)}, \boldsymbol{z}^{(j)})/\tau)}{\sum_{k=1}^{2N} \mathbf{1}_{[k \neq i]} \exp\big(\mathrm{sim}(\boldsymbol{z}^{(i)}, \boldsymbol{z}^{(k)})/\tau\big)}, \tag{2.5}$$

where $\tau > 0$ is a temperature parameter controlling the softmax sharpness, and $\mathbf{1}_{[k \neq i]} \in \{0, 1\}$ is an indicator excluding the self-similarity of the anchor $\boldsymbol{z}^{(i)}$ from the denominator.

The total loss is computed symmetrically over all positive pairs in the mini-batch. For each original example $\boldsymbol{x}^{(k)}$, indexed by $2k-1$ and $2k$ after augmentation, the full objective is defined as

$$\mathcal{L}_{\text{NT-Xent}} = \frac{1}{2N} \sum_{k=1}^{N} \Big[\ell(2k-1, 2k) + \ell(2k, 2k-1) \Big]. \tag{2.6}$$

The bidirectional formulation ensures that both views of each input contribute equally to the optimization process.

The original SimCLR approach demonstrated that carefully crafted augmentations, like color jitter for images, are essential to force models to disregard superficial cues and extract meaningful invariances. Yet, standard image augmentations are ill-suited for event data for two fundamental reasons. First, unlike conventional images, event streams lack absolute brightness values and color channels — the very characteristics that SimCLR's augmentations target. Second, event data encodes precise temporal dynamics and polarity changes that require specialized augmentations respecting their unique structure. Without such transformations, models may exploit temporal artifacts and learn shortcuts instead of developing true spatiotemporal understanding.

Our augmentations are crafted to preserve the spatiotemporal structure inherent in event streams while introducing variability that incentivizes the model to prioritize meaningful patterns rather than superficial correlations. The transformations can be grouped into spatial, temporal, and event-level perturbations. Below, we describe each of them:

1. **Random spatial shift**: Applies a uniform spatial displacement $(\Delta x, \Delta y)$ to all event coordinates, where $\Delta x \sim \mathcal{U}(-d_{max}, d_{max})$, with fixed $d_{max} \in [0, \min(W, H))$ and similarly for Δy. Events with transformed coordinates (x', y') satisfying $x' < 0$, $x' \geq W$, $y' < 0$, or $y' \geq H$ are eliminated to ensure all events remain within the valid sensor coordinates $[0, W) \times [0, H)$.

2. **Random horizontal flip**: Applies a horizontal mirror transformation to the event stream with probability p_{hflip} by flipping x-coordinates via $x' \leftarrow W - 1 - x$ across the sensor's vertical midline, while leaving y-coordinates, timestamps, and polarities unchanged. This guarantees valid sensor boundaries $x \in [0, W)$ and injects useful mirror symmetry into training data.

3. **Random temporal flip**: Reverses the sequence of events along the temporal axis with probability p_{tflip}, placing the last event first and the first event last. Timestamps are then adjusted to maintain relative timing by $t' \leftarrow \max(\boldsymbol{t}) - t$, where $\max(\boldsymbol{t})$ is the timestamp of the first event in the reversed sequence. To preserve consistent semantics of brightness changes, event polarities are simultaneously inverted using $p' \leftarrow -p$, thereby introducing time-reversal symmetry into the learned representations and producing physically plausible data.

4. **Random centering**: With probability p_{center}, performs spatiotemporal normalization by temporally shifting events as $t' \leftarrow t - \min(\boldsymbol{t})$ to align the first event at $t = 0$, and then spatially centering the event cloud via $x' \leftarrow x - \lfloor \frac{\min(\boldsymbol{x}) + \max(\boldsymbol{x}) + 1 - W}{2} \rfloor$ and $y' \leftarrow y - \lfloor \frac{\min(\boldsymbol{y}) + \max(\boldsymbol{y}) + 1 - H}{2} \rfloor$ to position the centroid at the sensor midpoint $(W/2, H/2)$. This modification drives the model to focus on the relative structure of events rather than their absolute position or timing.

5. **Random time window crop**: For each event stream selected with p_{tcrop}, it samples a duration ratio $\beta \sim \mathcal{U}(0, 1)$ and start ratio $\alpha \sim \mathcal{U}(0, 1 - \beta)$, retaining only events with timestamps $t \in \big[t_{\min} + \alpha \cdot (t_{\max} - t_{\min}),\ t_{\min} + (\alpha + \beta) \cdot (t_{\max} - t_{\min}) \big]$, where $t_{\min} = \min(\boldsymbol{t})$ and $t_{\max} = \max(\boldsymbol{t})$. This ensures that the full and cropped versions, capturing the same underlying object, are mapped closely in the embedding space. By varying the window size, the model avoids relying on shortcuts based on event sequence length and instead learns features invariant to temporal scaling. The approach is analogous to the global-local or adjacent view augmentations utilized in the original SimCLR paper.

6. **Temporal segment dropout**: Randomly selects a continuous temporal segment of fixed relative length $r \in [0, 1)$ and removes it from the event stream. Specifically, a start ratio $\alpha \sim \mathcal{U}(0, 1 - r)$ determines the interval $\big[t_{\min} + \alpha \cdot (t_{\max} - t_{\min}),\ t_{\min} + (\alpha + r) \cdot (t_{\max} - t_{\min}) \big]$, during which all events are discarded. The resulting stream retains events outside this interval, i.e., within $\big[t_{\min}, t_{\min} + \alpha \cdot (t_{\max} - t_{\min}) \big) \cup \big(t_{\min} + (\alpha + r) \cdot (t_{\max} - t_{\min}), t_{\max} \big]$. This requires the model to maintain consistent representations despite missing temporal segments, enhancing robustness to discontinuities and temporal occlusions in the event data.

7. **Spatial region dropout**: Samples a dropout square with side length $\sqrt{r} \cdot \min(W, H)$, where r is fixed relative area, centered at a point $(x_c, y_c) \sim \mathcal{U}(0, W-1) \times \mathcal{U}(0, H-1)$. All events falling within this region are discarded, simulating localized occlusions or sensor noise. This encourages the model to develop robust features that generalize despite partial loss of spatial information, preventing over-reliance on any specific zone of the sensor.

8. **Event-level dropout**: Independently removes each event with probability p_{drop} through Bernoulli sampling, resulting in a sparser set of events with original temporal order intact. This operation mimics the inherent noise of event-based sensors while the remaining events still reflect the underlying spatiotemporal patterns in the data.

9. **Gaussian noise**: Adds independent perturbations sampled from $\mathcal{N}(0, \sigma^2)$ to the spatial (x, y) coordinates and normalized timestamps of each event. After applying temporal offsets in the $[0, 1]$ range, the method rescales timestamps to the original interval and reorders events chronologically. It clamps spatial values to valid sensor bounds $[0, W-1] \times [0, H-1]$ and leaves polarity unchanged, simulating realistic distortions without altering the semantic structure of the event stream.

The event data augmentations described above are visualized in Figure 2.6.

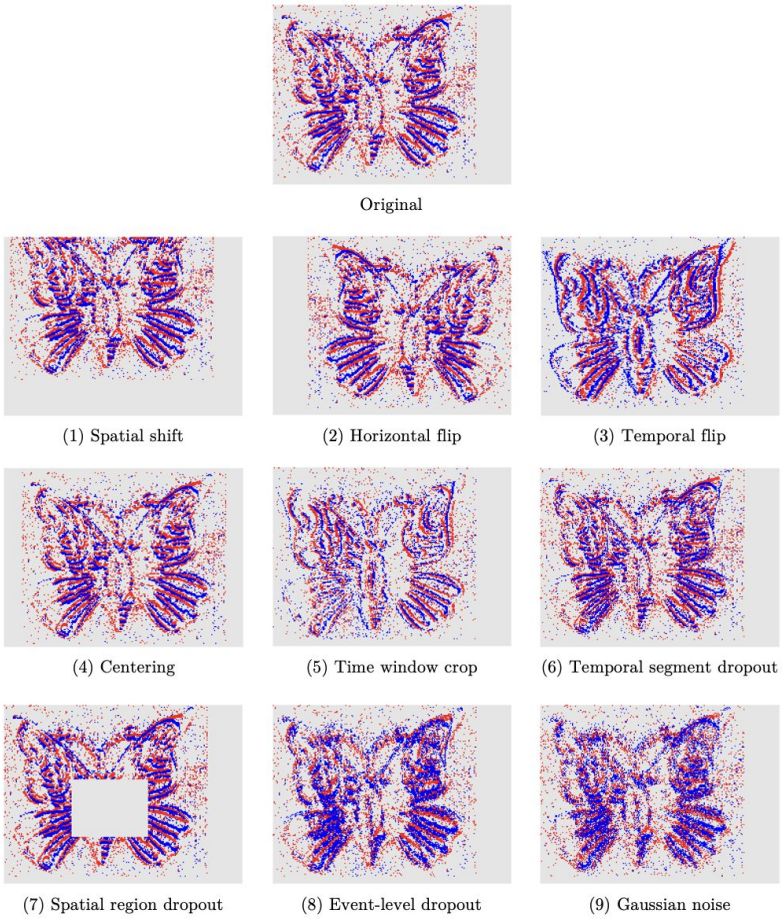

Figure 2.6: Event data augmentations.

In particular, temporal segment dropout, spatial region dropout, and event-level dropout form a group of drop-based transformations. Their design is motivated by EventDrop [29] work, which demonstrated the effectiveness of event elimination for improving generalization across various DNN architectures. These strategies are featured in Figure 2.7.

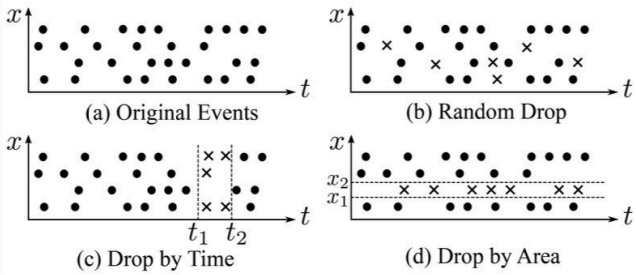

Figure 2.7: EventDrop augmentation. Source: [29].

The overall proposed framework is illustrated in Figure 2.8.

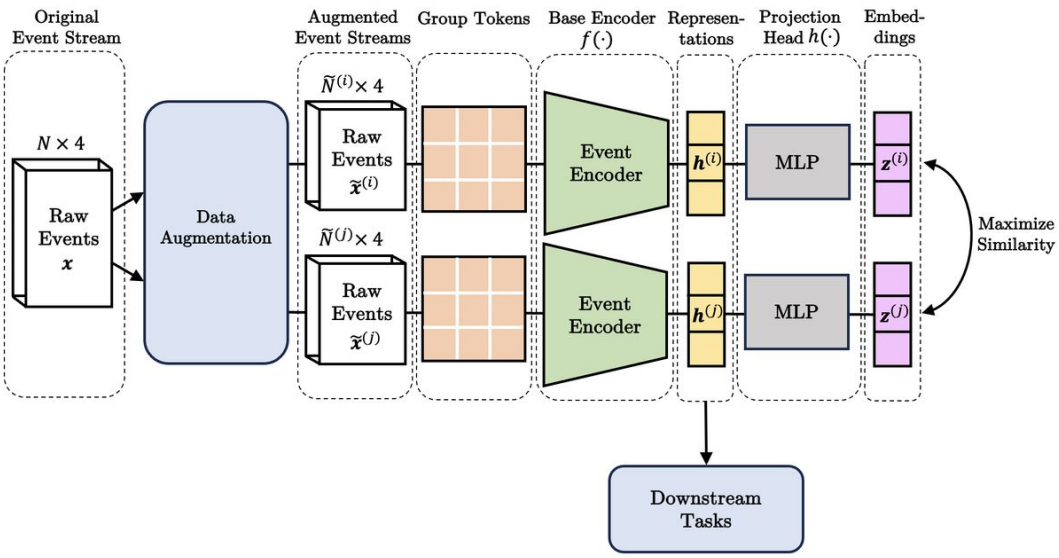

Figure 2.8: Unimodal contrastive pre-training.

Multimodal Contrastive Pre-training. To achieve effective multimodal comprehension, it is essential to bridge the representational gap between event-based visual input and natural language. To this end, we follow a learning strategy inspired by CLIP [55], which facilitates alignment between the two modalities in a shared latent space. We retain the event encoder previously trained on vision-specific contrastive task and couple it with a frozen CLIP text encoder. The training objective pulls together the embeddings of corresponding event and text pairs, while pushing apart those that are unrelated. This is implemented through a contrastive loss with two complementary components.

First, the event-to-text ($e2t$) loss ensures that each event embedding i is most similar to its paired description $j = i$, compared to other texts in the mini-batch. Let $\boldsymbol{e}^{(i)}$ and $\boldsymbol{t}^{(j)}$ denote the learned projections of the i-th event stream and j-th textual description, respectively. Formally, the $e2t$ objective is given by

$$\ell_{e2t}(i) = -\log \frac{\exp(\mathrm{sim}(\boldsymbol{e}^{(i)}, \boldsymbol{t}^{(i)})/\tau)}{\sum_{j=1}^{N} \exp(\mathrm{sim}(\boldsymbol{e}^{(i)}, \boldsymbol{t}^{(j)})/\tau)}, \tag{2.7}$$

where $\mathrm{sim}(\cdot, \cdot)$ stands for cosine similarity, $\tau > 0$ is a temperature scaling factor, and N is a mini-batch size.

Conversely, to ensure symmetric alignment, the text-to-event ($t2e$) loss encourages each $\boldsymbol{t}^{(j)}$ to attain the highest matching score relative to its associated $\boldsymbol{e}^{(j)}$. This is formulated as

$$\ell_{t2e}(j) = -\log \frac{\exp(\mathrm{sim}(\boldsymbol{e}^{(j)}, \boldsymbol{t}^{(j)})/\tau)}{\sum_{i=1}^{N} \exp(\mathrm{sim}(\boldsymbol{e}^{(i)}, \boldsymbol{t}^{(j)})/\tau)}. \tag{2.8}$$

The final training objective averages both directions across the batch to form a symmetric InfoNCE loss

$$\mathcal{L}_{\mathrm{InfoNCE}} = \frac{1}{2N} \left[\sum_{i=1}^{N} \ell_{e2t}(i) + \sum_{j=1}^{N} \ell_{t2e}(j) \right]. \tag{2.9}$$

The computation is implemented as shown below (Listing 2.1) in a CLIP-like setting. Here, event and text features are encoded separately, projected into a shared space, and compared using scaled cosine similarities. The final loss is computed symmetrically in both directions using cross-entropy over the similarity logits.

Listing 2.1: Numpy-style implementation of symmetric InfoNCE loss.

```
# event_encoder - Group Event Transformer
# text_encoder  - Text Transformer
# E[n, h, w, c] - minibatch of aligned group tokens
# T[n, l]       - minibatch of aligned texts
# W_e[d_e, d_h] - learned proj of event token to embedding space
# W_t[d_t, d_h] - learned proj of text to embedding space
# t             - learned temperature parameter

# extract feature representations of each modality
E_f = event_encoder(E) #[n, d_e]
T_f = text_encoder(T)  #[n, d_t]

# joint multimodal embedding [n, d_h]
E_h = l2_normalize(np.dot(E_f, W_e), axis=1)
T_h = l2_normalize(np.dot(T_f, W_t), axis=1)

# scaled pairwise cosine similarities [n, n]
logits = np.dot(E_h, T_h.T) * np.exp(t)

# symmetric loss function
labels = np.arange(n)
loss_e = cross_entropy_loss(logits, labels, axis=0)
loss_t = cross_entropy_loss(logits, labels, axis=1)
loss   = (loss_e + loss_t)/2
```

Figure 2.9 outlines our approach. Note that the text encoder remains frozen during training.

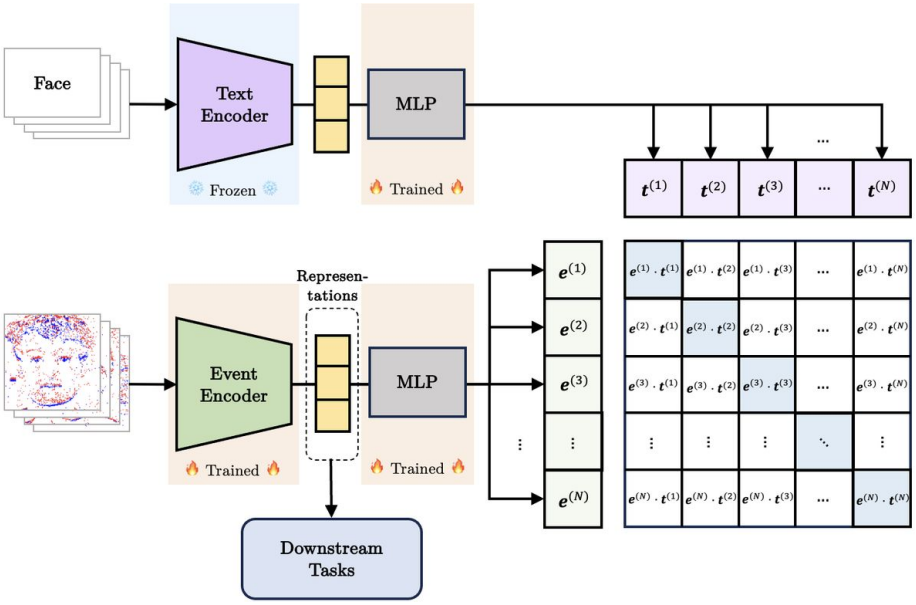

Figure 2.9: Multimodal contrastive pre-training.

Cross-Modal Finetuning. For cross-modal finetuning, we use our synthetic event-text dataset (Section 2.2), derived from a subset of InternVid. While the pre-training stages rely on real-world event data, finetuning is performed on simulated streams. This design choice is intended to encourage the model to generalize across both domains.

Our method integrates the BLIP [42] vision-grounded text decoder with the BLIP text encoder, as depicted in Figure 2.10. A primary component of the training objective is based on next-word prediction, where the model incrementally learns to generate coherent text sequences in an autoregressive fashion. At each timestep, the model estimates the conditional distribution of the next token given the preceding tokens and the event embedding.

Formally, provided an input token sequence $x_{1:T}$ and a learned event representation $\boldsymbol{e}$, the autoregressive likelihood is defined as

$$p_\theta(x_{1:T} \mid \boldsymbol{e}) = \prod_{t=1}^{T} p_\theta(x_t \mid x_{<t}, \boldsymbol{e}), \tag{2.10}$$

where $p_\theta(x_t \mid x_{<t}, \boldsymbol{e})$ is the probability of producing token x_t conditioned on the preceding context $x_{<t}$ and event feature $\boldsymbol{e}$. This factorization underscores the sequential nature of the decoder, which relies on causal masking to ensure that predictions depend exclusively on the history established by earlier tokens.

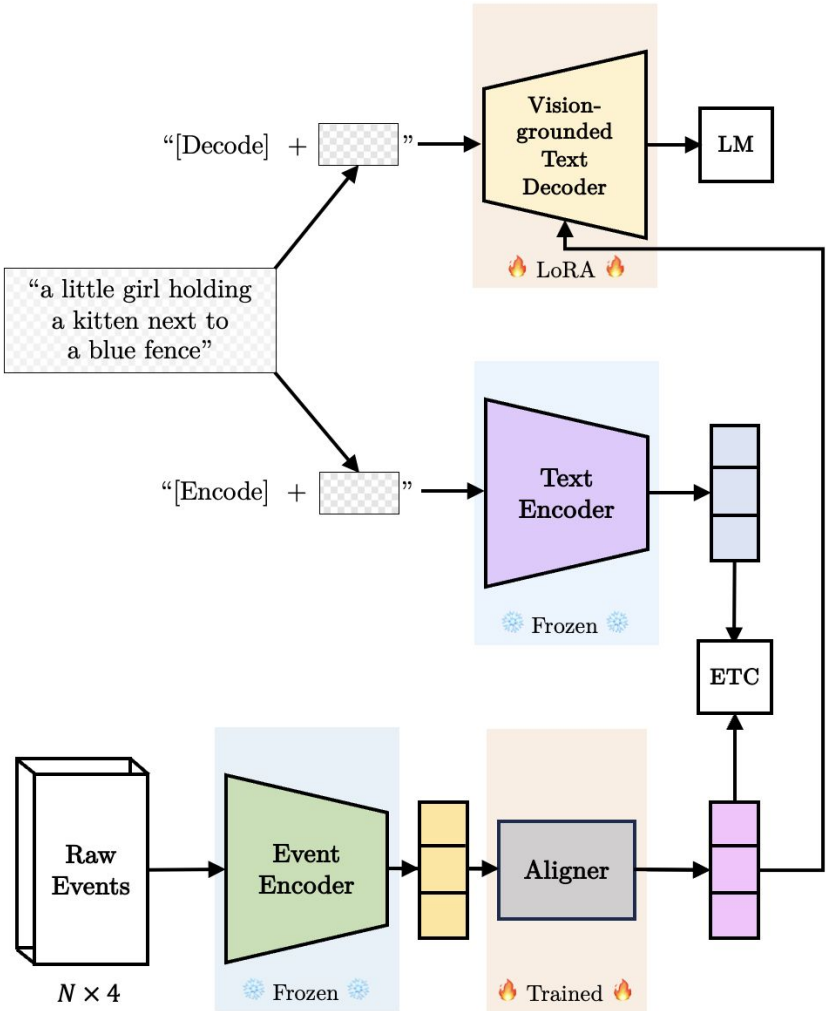

Figure 2.10: Cross-modal finetuning.

The language modeling loss is then defined as the negative log-likelihood

$$\mathcal{L}_{LM} = -\frac{1}{\sum_{i=1}^{N} T_i} \sum_{i=1}^{N} \sum_{t=1}^{T_i} \log p_\theta(x_t^{(i)} \mid x_{<t}^{(i)}, \boldsymbol{e}^{(i)}), \tag{2.11}$$

where T_i is the number of non-padding tokens in sequence i. By construction, it penalizes deviations from the ground-truth sequence, guiding the model toward linguistically consistent and modality-aligned text.

If training relies solely on the LM objective, the model risks representational collapse, a degenerate case where embeddings from different inputs converge to nearly identical, uninformative vectors. The decoder achieves low loss by exploiting shortcuts in the latent representation, thereby ignoring structural information encoded in the event features. To avoid this, we incorporate an additional InfoNCE objective, introduced in multimodal pre-training, referred to here as event-text contrastive (ETC) in line with BLIP terminology.

The total training loss combines the LM and ETC parts:

$$\mathcal{L}_{total} = \mathcal{L}_{LM} + \lambda \cdot \mathcal{L}_{ETC}, \tag{2.12}$$

where $\lambda > 0$ controls the trade-off between generative performance and cross-modal alignment.

To retain the expressive power of the pre-trained event encoder, its parameters are kept frozen throughout finetuning process. Optimization is restricted to the aligner module, which projects event features into the decoder's embedding space, and to the decoder parameters, selectively adapted using LoRA.

LoRA (Low-Rank Adaptation) [31] is a parameter-efficient finetuning technique that injects trainable low-rank matrices into the dense layers of a transformer, allowing targeted updates without modifying the original weights. Rather than learning a full-rank parameter matrix, LoRA approximates $\Delta \boldsymbol{W}$ through a low-rank decomposition, significantly reducing the number of trainable parameters while maintaining modeling capacity. Precisely, the corrected weights are specified as

$$\boldsymbol{W}_{\text{updated}} \leftarrow \boldsymbol{W}_{\text{pre-trained}} + \boldsymbol{B}\boldsymbol{A}, \tag{2.13}$$

where $\boldsymbol{W}_{\text{pre-trained}} \in \mathbb{R}^{d \times k}$ and $\boldsymbol{B} \in \mathbb{R}^{d \times r}$, $\boldsymbol{A} \in \mathbb{R}^{r \times k}$ are low-rank matrices with $r \ll \min(d, k)$. The adaptation rank $r \in \mathbb{N}$ is chosen to balance computational efficiency and model performance.

By constraining parameter updates to low-rank perturbations, LoRA preserves the pre-trained model's inductive biases while enabling scalable adaptation. This approach is particularly well-suited for finetuning large-scale models under resource constraints.

Chapter 3

Experiments

The chapter is organized as follows: Section 3.1 presents the datasets utilized in our experiments, Section 3.2 reports the initial trials conducted in the Event-CLIP framework, Section 3.3 outlines a proof-of-concept for our proposed unimodal pre-training strategy, and Section 3.4 describes the experiments leading to the event-based vision-language model for event stream captioning.

3.1 Datasets

In this section, we give a concise overview of the neuromorphic datasets used in our experiments.

N-Caltech101 [50] is an event-based equivalent of the Caltech101 [20] image dataset, created by moving an event camera around a monitor displaying static Caltech101 images. It covers 101 distinct classes, including animals, vehicles, and household items, with a total of approximately 8,700 samples.

CIFAR10-DVS [39] is a neuromorphic adaptation of the widely used CIFAR-10 [38] image classification dataset. It was generated by displaying CIFAR-10 images on a monitor while simulating translational motion, which was then recorded using a static DVS camera. The dataset contains 10 object categories such as airplanes, cats, and trucks, making it a standard benchmark for event-based classification.

N-CARS [62] represents the first real-world event-based vehicle detection benchmark. It was collected using an ATIS mounted on a moving car navigating urban environments. Unlike synthetic datasets, N-CARS inherently contains motion-induced noise such as vehicle vibrations, resulting in sparse and noisy event streams that better reflect real-world neuromorphic data. The dataset includes only two classes: cars and non-cars, with a total of 12,336 car samples and 11,693 non-car samples.

N-ImageNet Mini [36]. N-ImageNet is a neuromorphic version of the large-scale ImageNet [13] dataset, created using a methodology similar to N-Caltech101. The complete dataset comprises over 1.7 million event sequences spanning 1000

classes. Despite the large number of samples, the short duration of each recording limits the presence of meaningful motion cues, as highlighted in [33]. Consequently, N-ImageNet is not suitable as a standalone resource for event-based scene understanding, which relies heavily on rich temporal dynamics. Thus, we employ the Mini variant containing approximately 135,000 samples across 100 classes alongside other complementary datasets to ensure adequate representation of event-based information.

DailyDVS-200 [67] is an extensive event-based dataset, covering 200 action categories in real-world scenarios. Recorded by 47 participants, it features more than 22,000 event streams, capturing diverse action types, scene complexities, and acquisition conditions. Each sequence is meticulously annotated with 14 distinct attributes, enabling fine-grained analysis and characterization of the recorded actions.

Table 3.1 summarizes the specifications of the aforementioned datasets. The *Scale* column indicates the total sample count, while *Duration* shows the maximum event stream length (in milliseconds) for each dataset.

Dataset	**Year**	**Sensor**	**W × H**	**Scale**	**Classes**	**Duration (ms)**
N-Caltech101	2015	ATIS	240 × 180	8,709	101	300
CIFAR10-DVS	2017	DAVIS128	128 × 128	10,000	10	1,200
N-CARS	2018	ATIS	120 × 100	24,029	2	100
N-ImageNet Mini	2021	Samsung DVS Gen3	640 × 480	134,395	100	50
DailyDVS-200	2024	DVXplorer Lite	320 × 240	22,046	200	20,000

Table 3.1: Comparison of public event datasets used in our work.

3.2 Initial Experiments

Our initial focus was on integrating the DeepSeek-VL-1.3B-base [48] backbone into the EventCLIP [73] framework, motivated by the model's recent open-source release and its promising performance on vision-grounded text generation tasks like VQA and image captioning. All experiments were conducted using the N-Caltech101 dataset.

Experiment 1: Zero-Shot Classification. The first experiment evaluated whether replacing CLIP's backbone with DeepSeek-VL could improve zero-shot classification. However, unlike CLIP, which was pre-trained with a cosine similarity-based objective, DeepSeek-VL aligns vision and language through next token prediction loss. Hence, its zero-shot classification on the N-Caltech101 dataset matched random guessing, with approximately 1% accuracy across 101 classes. These findings indicate that the absence of a contrastive alignment in DeepSeek-VL limits its capacity for effective CLIP-style zero-shot transfer to classification tasks.

Experiment 2: Contrastive Finetuning. In the next experiment, we finetuned alignment module of DeepSeek-VL together with text features from its language component. Training was performed using a cosine decay schedule with warmup over 50 epochs, ultimately reaching **91.67%** validation accuracy.

The model showed particularly fast convergence, achieving 86.56% accuracy after just one epoch. The observations suggest that while DeepSeek-VL does not share CLIP's inherent zero-shot classification capabilities, its vision-language alignment can be successfully adapted to specific tasks through finetuning.

Experiment 3: Hyperparameter Search. The third experiment applied a Bayesian hyperparameter sweep using Weights & Biases (WandB) [5] to jointly finetune the aligner and text features. The sweep frequently favored the prompt "`an edge map of a {label}`" among tested configurations, achieving a top accuracy of **91.10%**. While the automated hyperparameter search fell slightly short of manual tuning results in the previous experiment, it provided valuable insights into suitable prompt configurations and highlighted promising directions for further optimization.

Experiment 4: Finetuning on Reconstructed Video Frames. As a baseline comparison, the fourth experiment finetuned the aligner and text features on video frames reconstructed from raw event streams using E2VID [58] (Figure 3.1). This approach reached **94.60%** classification accuracy, outperforming the 91.67% obtained with event histograms. Given that DeepSeek-VL was pretrained on natural images, its stronger results on video frames were anticipated. Nevertheless, the modest performance gap reveals that DeepSeek-VL adapts well to event-based representations, even when processing highly compressed inputs like event histograms.

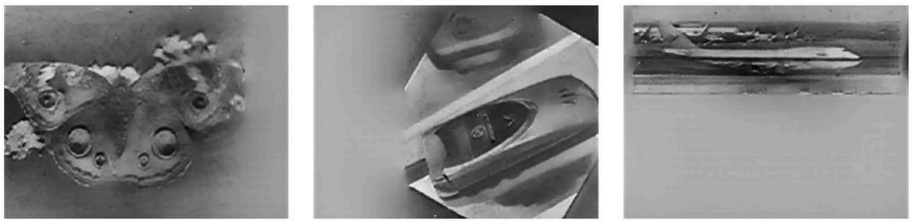

Figure 3.1: Video frames reconstructed from N-Caltech101 with E2VID.

3.3 Pre-training Study

This part provides a preliminary evaluation of the proposed unimodal contrastive pre-training (described in Section 2.3), serving as a proof of concept, and subsequently examines the cross-modal finetuning. The evaluation employs a 3-stage GET network as our prototype encoder design, following the implementation from the original paper [53]. GET encoder, configured as specified in Table 3.2, contains a total of 4.5 million parameters. Its architecture is illustrated in Figure 3.2. For this setup, we used CIFAR10-DVS dataset.

Unimodal Contrastive Pre-training. Guided by the insights of SimCLR [10], we explored the role of batch size in effective contrastive learning. The study emphasizes that larger batch sizes offer a substantial number of negative samples, which significantly improves representation quality. Therefore, we experimented with batch sizes of 256, 512, and 1024.

Patch size	4
Embedding dimension	48
Depths	[2, 2, 8]
Number of heads	[3, 6, 12]
Window size	8
MLP Ratio	4
Dropout probability	0.2
Group number	12

Table 3.2: Prototype GET configuration.

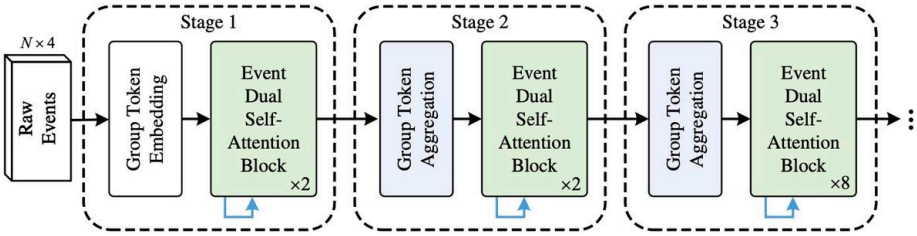

Figure 3.2: Prototype GET architecture. Source: [53]

In [10], training was performed on a single TPU, allowing for a maximum batch size of 4096. In our case, such computational resources were not available, and storing large event sequences imposed substantial memory demands. Our setup consisted of NVIDIA RTX 8000 GPUs with 50 GB of VRAM each. To reach a batch size of 1024, we used three GPUs and adapted the NT-Xent loss function for distributed training. Since direct sample gathering across devices is not inherently differentiable, we used PyTorch Lightning's [19] implementation of differentiable cross-device tensor gathering, given in Listing 3.1.

Listing 3.1: Differentiable cross-device tensor gathering. Source: [19]

```
class SyncFunction(torch.autograd.Function):
    """
    Synchronizes tensors across all processes in distributed training.
    """

    @staticmethod
    def forward(ctx,
                tensor: torch.Tensor) -> torch.Tensor:
        """
        Gathers tensors from all processes and returns concatenated result.

        Args:
            ctx: Context object to save information for backward pass
            tensor (torch.Tensor): Input tensor to synchronize

        Returns:
            torch.Tensor: Concatenated tensors from all processes
        """

        if not dist.is_initialized():
            return tensor
```

```
        ctx.batch_size = tensor.size(0)
        world_size = dist.get_world_size()

        # Validate input tensor
        if not tensor.is_contiguous():
            tensor = tensor.contiguous()

        # Pre-allocate gather list with proper device placement
        gathered_tensors = [torch.empty_like(tensor)
                            for _ in range(world_size)]

        # Perform all-gather
        dist.all_gather(gathered_tensors, tensor)

        # Concatenate results
        return torch.cat(gathered_tensors, dim=0)

    @staticmethod
    def backward(ctx,
                 grad_output: torch.Tensor) -> torch.Tensor:
        """
        Scatters gradients back to original processes.

        Args:
            ctx: Context object to save information for backward pass
            grad_output (torch.Tensor): Gradient of the concatenated output

        Returns:
            torch.Tensor: Gradient for the original input tensor
        """

        if not dist.is_initialized():
            return grad_output

        # Validate gradient tensor
        if not grad_output.is_contiguous():
            grad_output = grad_output.contiguous()

        rank = dist.get_rank()
        batch_size = ctx.batch_size

        # Reduce gradients across all processes (sum)
        grad_input = grad_output.clone()
        dist.all_reduce(grad_input, op=dist.ReduceOp.SUM, async_op=False)

        # Calculate slice indices for this process
        start_idx = rank * batch_size
        end_idx = start_idx + batch_size

        # Handle incomplete batches (if any)
        end_idx = min(end_idx, grad_input.size(0))

        # Get gradient slice for this process
        return grad_input[start_idx:end_idx]
```

The `SyncFunction` gathers feature tensors from all GPUs during the forward pass and, in the backward pass, reduces the gradients and distributes them back to the corresponding devices. This enables the contrastive loss to operate on a global batch spanning all GPUs while preserving the correct gradient flow.

To assess the quality of the learned representations, we applied a linear probe by training a multi-class logistic regression classifier on features extracted from the training set while keeping the encoder frozen. The classifier's accuracy was then measured on the validation set. Strong performance with this simple model suggests that the learned embeddings are highly informative.

Our results demonstrate that linear probing accuracy improves with larger batch sizes, as depicted in Figure 3.3 (a), attaining a maximum of **60.8%**. Training was conducted for 200 epochs, and as shown in Figure 3.3 (b), accuracy increased steadily over time for all batch sizes. This aligns with the findings of SimCLR, which reported that longer training durations lead to more robust representations.

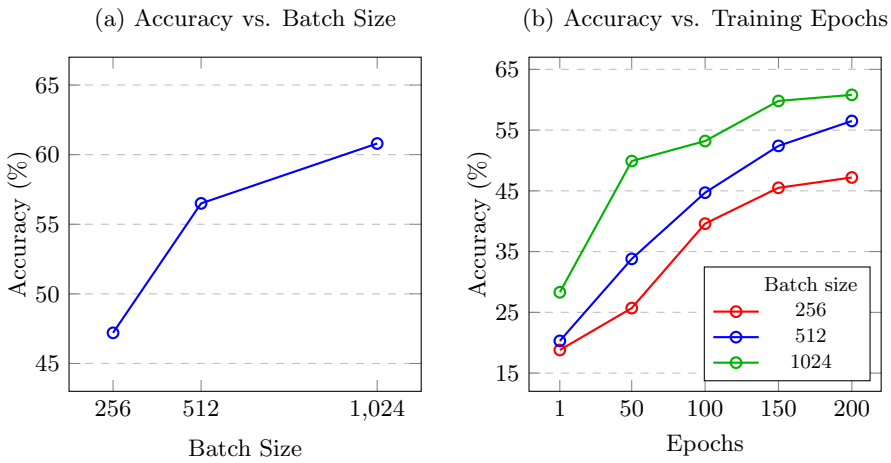

Figure 3.3: Linear probe accuracy on CIFAR10-DVS: (a) as function of batch size and (b) training epochs.

Crucially, our unimodal pre-training stage provides a strong initialization for subsequent finetuning, with the batch-size-1024 variant standing out. When finetuned end-to-end with a classification head, it achieves **79.7%** accuracy despite having only 4.5 million parameters. This surpasses previous pre-training approaches (78.0% by [74] and 78.6% by [33]), which rely on models three to five times larger, underscoring the data efficiency of our method. For reference, the fully supervised GET implementation reaches 84.8%.

We aim to compare the encoder representations across three training strategies: (1) fully supervised, (2) pre-trained, and (3) finetuned. For each setting, features from the validation set were extracted using the corresponding frozen backbone and then processed with t-SNE. t-SNE stands for *t-distributed stochastic neighbor embedding* and is an unsupervised dimensionality reduction technique. In our case, the encoder produces 192-dimensional feature vectors, which t-SNE maps into two dimensions for subsequent visualization in Figure 3.4.

In the fully supervised scenario, embeddings form distinct clusters with moderate interclass separation, though some boundaries remain fuzzy. This yields the highest classification accuracy, but reflects strong optimization for the specific task and limited generality. The pre-trained model exhibits significant overlap

among classes, indicating limited discriminative capacity prior to task-specific training.

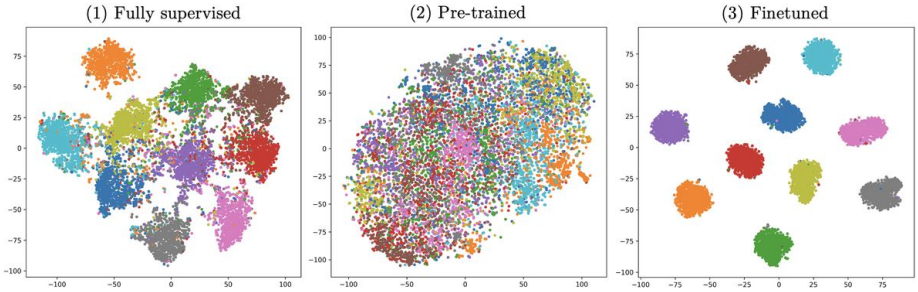

Figure 3.4: t-SNE of frozen GET embeddings from CIFAR10-DVS.

After finetuning, the embeddings form tightly packed and clearly separated clusters, confirming that supervised adaptation of a generic model refines into a more structured and class-distinctive representation space. However, the accuracy of the finetuned model on CIFAR10-DVS is slightly lower than under full supervision, highlighting that better spatial arrangement does not necessarily translate to superior task performance.

Cross-Modal Finetuning. Our experiments involved two different aligners: one based on MLP initialized randomly, and the other pre-trained on image-text pairs from DeepSeek-VL. The LLM was also obtained from DeepSeek-VL, but unlike the standard DeepSeek LLM, this variant was initialized from an intermediate checkpoint of the pre-trained model and then further finetuned on image-text pairs [48]. Such adaptation enhanced the LLM's ability to associate visual and textual information, allowing it to handle multimodal tasks more effectively.

The event encoder, pre-trained in SimCLR-style, and aligner were finetuned using a text-augmented CIFAR10-DVS classification dataset. Due to the absence of a dedicated event-text dataset during this study, we extended the original CIFAR10-DVS labels with comprehensive textual descriptions to enhance language-relevant supervision. For example, the label "automobile" was supplemented with alternative phrasings such as "a vehicle with four wheels" and "a car driving on the road". We also ensured diverse query prompts such as "What is captured by the event stream?", "Describe the content of the event data", and other semantically similar variations.

The LLM remained frozen, while the projected event embeddings were prepended to the text query embeddings as soft visual prompts, conditioning the LLM on the event-based visual representations (Figure 3.5). However, this architecture eventually led to overfitting, which we attribute to two primary factors: (1) the capacity mismatch between the relatively small encoder and the 1B-parameter LLM, and (2) the finite variability of class descriptions. This limitation became clearly visible when the prototype VLM, at its optimal training stage, started producing hallucinated text, often displaying meaningless repetitions of class labels. Based on these observations, we adopted a larger encoder architecture,

added vision-language alignment, and constructed an event-text dataset for our main experiments.

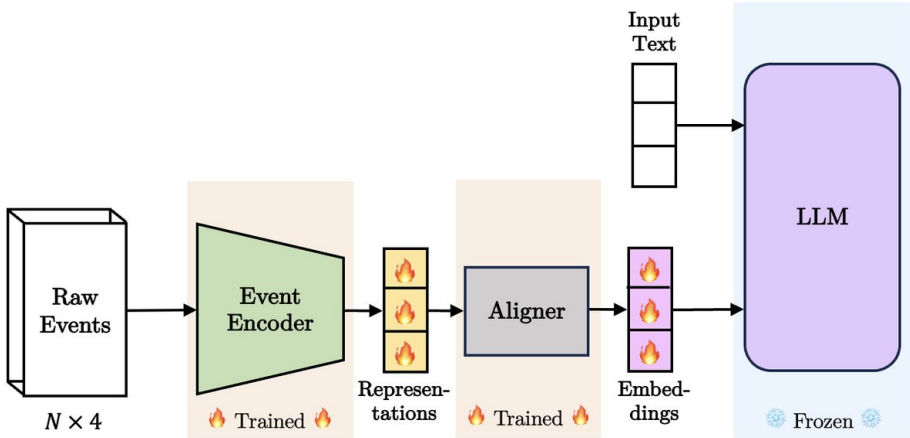

Figure 3.5: Prototype VLM architecture.

3.4 Experimental Setup

The GET encoder, configured as specified in Table 3.3, contains a total of 17 million parameters. Its architecture is illustrated in Figure 3.6.

Patch size	4
Embedding dimension	48
Depths	[2, 2, 8, 8]
Number of heads	[3, 6, 12, 24]
Window size	8
MLP Ratio	3
Dropout probability	0.2
Group number	12

Table 3.3: GET configuration.

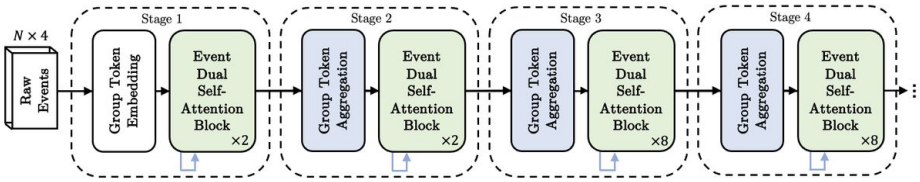

Figure 3.6: GET architecture.

The encoder was pre-trained on the combined datasets presented in Section 3.1. Since the sample counts in Table 3.1 include training, validation, and occasionally test splits, the final training set of the aggregated datasets contained over 172,000 samples (Figure 3.7), complemented by more than 21,000 validation examples. Test splits were excluded entirely, as they were not utilized in any

stage of model development or evaluation. The merged dataset encompassed 410 classes; three duplicate classes were identified across datasets and merged into single categories. To ensure consistency, all events were resized to a reference resolution of 240×240 prior to tokenization, as the GET group tokenizer requires a fixed sensor resolution for spatial partitioning.

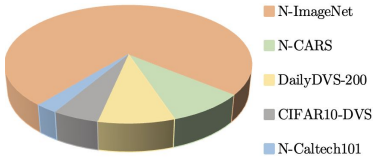

Figure 3.7: Proportional distribution of datasets in the pre-training corpus.

Unimodal Contrastive Pre-training. The augmentation hyperparameters, detailed alongside the augmentation methodology in Section 2.3, were configured according to Table 3.4, which are identical to those used in the proof-of-concept experiments.

d_{max}	50
p_{hflip}	0.5
p_{tflip}	0.8
p_{center}	0.5
p_{drop}	0.1
r	0.1
σ	0.1

Table 3.4: Augmentation hyperparameters.

Recognizing that the computational demands, driven by large batch sizes, dataset scale, and increased model complexity, exceeded our initial resources, we secured access to high-performance infrastructure through the Swiss AI Initiative. As the largest open science effort for AI foundation models globally, this is the first initiative of the Swiss National AI Institute, a collaboration between the ETH AI Center and the EPFL AI Center. Leveraging this support, we conducted pre-training on a node of 4 NVIDIA GH200 GPUs, each equipped with 120GB of VRAM.

Dataset	**Classes**	**Unimodal Acc.** (%)	**Multimodal Acc.** (%)	**Δ** (%)
N-Caltech101	101	51.3	60.8	+9.5
CIFAR10-DVS	10	48.6	83.9	+35.3
N-CARS	2	89.6	93.7	+4.1
N-ImageNet Mini	100	19.1	27.8	+8.7
DailyDVS-200	200	7.4	14.2	+6.8

Table 3.5: Linear probe accuracy on public datasets.

Since the pre-training procedure included mapping randomly augmented variants of the same input to closely aligned representations within the embedding space, the encoder can be effectively adapted for event-to-event retrieval tasks. In this framework, given a reference event stream, the objective is to retrieve

semantically similar sequences from a dataset by leveraging the learned embedding structure. To qualitatively assess the retrieval capabilities of the encoder, we used it to extract the most similar validation set matches for selected training examples. The corresponding pairings appear in Figure 3.8.

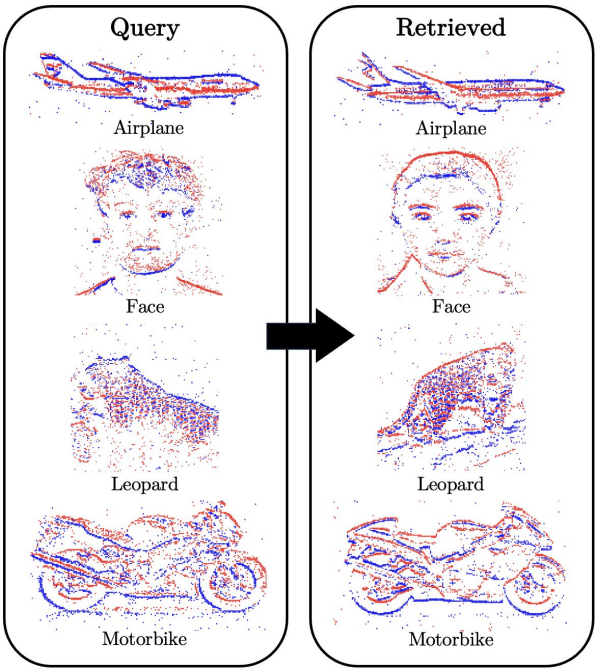

Figure 3.8: Event-to-event retrieval.

To evaluate retrieval performance, we computed the mean Average Precision at K ($mAP@K$), where K denotes the maximum rank examined. Query features were extracted from the training set, and gallery features from the validation set, ensuring no overlap between them. For each query q, we calculated similarity scores with all gallery items and ranked the results. Here, all items within the gallery dataset that share the same class label as the query are considered relevant.

The Average Precision at K for a single query q is given by

$$AP_K(q) = \frac{1}{R_q} \sum_{k=1}^{K} P_q(k) \cdot rel_q(k), \tag{3.1}$$

where R_q is the total number of relevant items, $P_q(k)$ is the precision computed at rank k, and $rel_q(k)$ is a relevance indicator equal to 1 if the item at rank k is in the same class as q, and 0 otherwise.

The mean Average Precision at K ($mAP@K$) is then

$$mAP@K = \frac{1}{M} \sum_{q=1}^{M} AP_K(q), \tag{3.2}$$

where M is the number of queries.

Figure 3.9 reports the $mAP@K$ for the encoder under unimodal and multimodal pre-training.

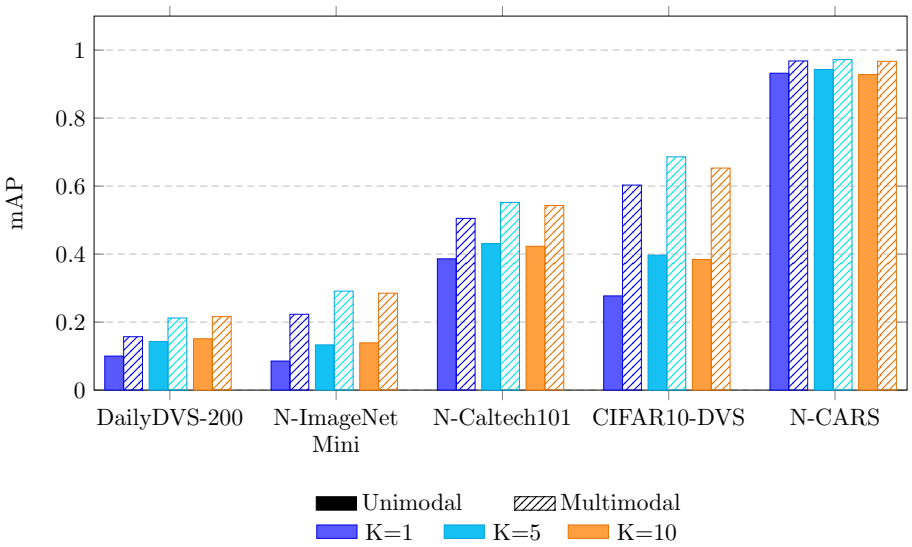

Figure 3.9: $mAP@K$ across public datasets.

Multimodal Contrastive Pre-training. To retain the low-level features learned during the initial unimodal pre-training, we freeze the first stage of the encoder while training the remaining three stages in the multimodal contrastive regime.

Because the combined dataset contains label information that was not utilized in the SimCLR-inspired setting, we leverage the textual form of these labels, more precisely, the prompt "`an event stream corresponding to {label}`" to enable event-text alignment, which is absent in the prototype experiments. We posit that the second phase of pre-training not only bridges the representational gap between event-based vision and language, but also refines the encoder's latent space, thereby enhancing its suitability for downstream tasks. As shown in Table 3.5, applying a multimodal objective following unimodal contrastive pre-training increases linear probe accuracies across all used public datasets.

Figure 3.9 illustrates that for $K \in \{1, 5, 10\}$, the multimodally trained model achieves consistently higher $mAP@K$ across all datasets and ranks K compared to the unimodally pre-trained version. The largest improvement is observed on CIFAR10-DVS, where the unimodal variant falls notably short. For both pre-training phases, $mAP@K$ increases with K, indicating that relevant matches are often retrieved beyond the top-1 position. On N-CARS, a binary classification dataset, both methods reach near-perfect scores, reflecting the relative simplicity of this task.

We visualize the resulting embeddings following the approach in Section 3.3. For clarity, we restrict our analysis to CIFAR10-DVS and N-CARS, as the other datasets contain too many classes for meaningful visualization. In Figure 3.10 (a), compared to Figure 3.4 (2) from our prototype, the class separation for CIFAR10-DVS is noticeably clearer. For the N-CARS dataset in Figure 3.10 (b), the separation is nearly perfect.

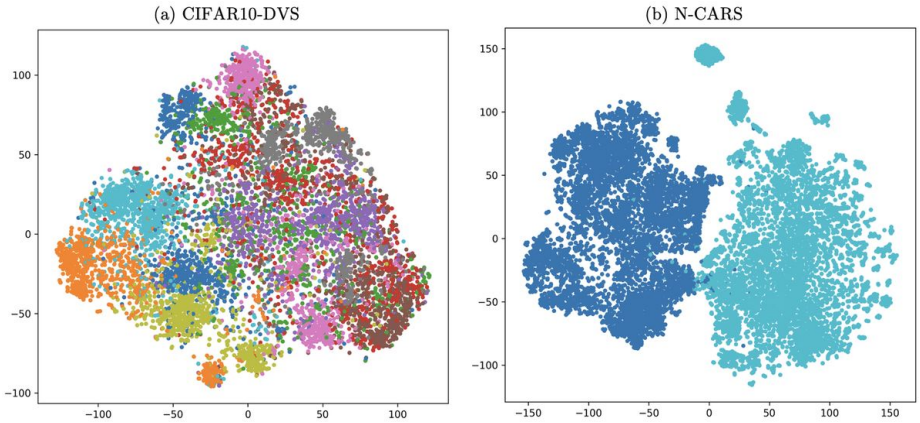

Figure 3.10: t-SNE of frozen GET embeddings.

From a theoretical perspective, the projection head mapping to the joint vision-language space could, in principle, attain strong performance merely by aligning pre-existing feature structures with text embeddings, without enhancing the underlying representations. However, our results demonstrate that the encoder develops significantly richer embeddings, thereby improving the structure of the latent space.

Similar to event-to-event, we qualitatively assessed text-to-event retrieval, examining the model's ability to identify relevant event sequences from textual prompts. Instead of adhering strictly to the training prompt format "`an event stream corresponding to {label}`", we introduced alternative formulations to probe robustness. Specifically, we tried synonymous expressions for "event stream" (e.g. "neuromorphic recording", "asynchronous signal"), employed a more general phrasing such as "`this is {label}`", and also substituted the class label with related terminology.

As shown in Figure 3.11, the model successfully retrieved event sequences from the correct classes across all prompt variations. While the first three expressions explicitly contained the ground-truth label, the final prompt "`money`" still returned a matching sample from the "dollar bill" class. This suggests that the model can generalize beyond its training prompt template, demonstrating sensitivity to semantically related terms in retrieval.

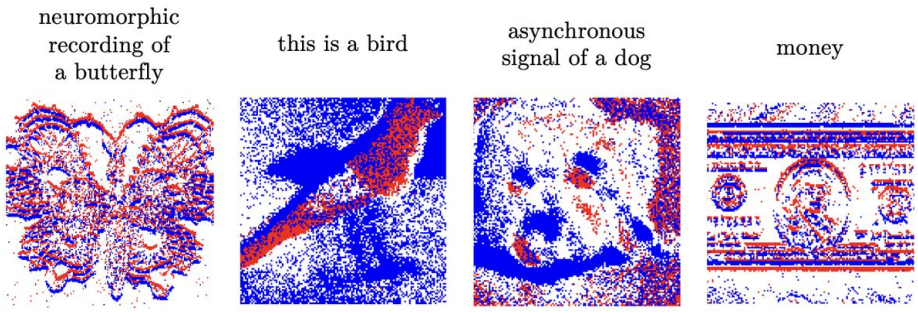

Figure 3.11: Text-to-event retrieval.

Overall, the two pre-training phases yield a transformer encoder that operates on event data without converting it into frame-based inputs, equipped with general knowledge of event structures of varying lengths and characteristics across datasets, and ready for downstream tasks.

Cross-Modal Finetuning was conducted as described in Section 2.3. We applied LoRA with rank 8 on the query and value matrices of the BLIP text decoder, yielding approximately 600,000 parameters ($\approx 0.4\%$ of the trainable parameters of the full decoder). Our constructed event-text dataset (Section 2.2) was randomly partitioned into 91,800 training pairs and 10,200 validation pairs (a 90%/10% split). To preserve event semantics while encouraging generalization, we incorporated random spatial shifts, random horizontal flips, and event-level dropout augmentations. As the dataset does not include questions, we focused exclusively on the task of **event stream captioning**. However, the model could, in principle, be adapted for event-based question answering via prompting or finetuning on event-question-answer triplets.

We report results using BLEU [52] and BERTScore [78], two common automatic text evaluation metrics. BLEU measures n-gram overlap between generated and reference captions, providing a fast and reproducible comparison across models. Its limitations, however, are well-documented: it penalizes synonyms, paraphrases, and semantically correct but lexically different sentences. To complement this, we employed BERTScore, which measures semantic similarity between tokens by embedding them in a BERT [17] space and computing cosine similarity. Given that BLIP itself uses a BERT-based architecture for its text encoder, this criterion is particularly well-suited for evaluating our model's outputs.

For caption generation, we employed greedy decoding, where the model deterministically selects the most probable token at each step. This method was chosen over the higher computational cost of beam search or the output variance of sampling techniques to ensure that the reported metrics offer a deterministic and reproducible evaluation of model performance.

Metric	**Score**
BLEU-1	0.13
BLEU-2	0.06
BLEU-4	0.02
BERTScore	0.17

Table 3.6: Evaluation results on the neuromorphic InternVid subset.

The metric values, shown in Table 3.6, are relatively low, which can be attributed both to the modest model size and the inherent difficulty of the dataset. BLEU-n denotes BLEU scores calculated with n-gram precision, where higher n corresponds to longer matching word sequences. Nevertheless, the automatic scores alone do not fully convey the strengths of our model, as demonstrated by the qualitative examples below.

Frames:	
GT:	The field in the gym has white cones
Pred:	A man is playing soccer in a gym
Scores:	BLEU-1: 0.25 | BLEU-2: 0.13 | BLEU-4: 0.07 | BERT: 0.35
Frames:	
GT:	A hand holding an item that is wrapped in plastic
Pred:	A person is holding a small piece of plastic
Scores:	BLEU-1: 0.40 | BLEU-2: 0.15 | BLEU-4: 0.06 | BERT: 0.53
Frames:	
GT:	Wakeboarder catches air in the lake during the summer of 2013
Pred:	A man in the ocean with a surfboard big waves
Scores:	BLEU-1: 0.18 | BLEU-2: 0.13 | BLEU-4: 0.08 | BERT: 0.14

Figure 3.12: Success modes of event stream captioning.

Figure 3.12 contains several examples from the validation set in which the generated captions align successfully with the visual content. In the first case, the event stream distinctly captures the player as the central figure of the scene, while the cones in the background are only incidental. Our produced description emphasizes the man, correctly characterizing him as "playing soccer in a gym". In contrast, the ground truth annotation refers to the cones, which do not represent the actual focus of the sequence.

In the second example, although the model's caption differs lexically from the ground truth ("person" vs. "hand", "piece of plastic" vs. "wrapped in plastic"), the semantic meaning remains consistent with the recorded action. Despite this alignment, the automatic evaluation fails to recognize it, yielding a disproportionately low BERTScore of 0.53.

The third case illustrates a more challenging scenario: while the ground truth specifies a lake and a wakeboard, the model predicts "ocean" and "surfboard". Differentiating between these concepts often requires domain-specific expertise and fine-grained visual analysis from a viewer, making our description still rea-

sonably accurate. Finally, details such as the recording year lie outside the objectives of caption generation and are consequently absent from the model's outputs.

Overall, a clear discrepancy emerges between human judgment and automated evaluation. While the generated captions adequately describe event data, widely used metrics such as BLEU and BERTScore are unable to capture this correspondence. Because they operate purely on linguistic similarity without taking the visual context into account, they tend to undervalue conceptually valid outputs. For instance, the third caption, though reasonable in meaning, receives a score of no more than 0.2 from both criteria.

Frames:	
GT:	A man is holding up a soccer ball on a field
Pred:	A man is holding a small dog in his arms
Scores:	BLEU-1: 0.45 | BLEU-2: 0.36 | BLEU-4: 0.25 | BERT: 0.55
Frames:	
GT:	Yahoo finance live 2019
Pred:	A black background with the word 'subscribe' on it
Scores:	BLEU-1: 0.00 | BLEU-2: 0.00 | BLEU-4: 0.00 | BERT: -0.17
Frames:	
GT:	A woman with her bike is taking a selfie
Pred:	A man in a yellow shirt is standing in a field
Scores:	BLEU-1: 0.27 | BLEU-2: 0.12 | BLEU-4: 0.05 | BERT: 0.38

Figure 3.13: Failure modes of event stream captioning.

The second set highlights typical errors. A common failure is capturing the action but misidentifying the object. In the first case, the model described "a small dog" instead of the soccer ball actually being held. Such mistakes often do not drastically lower the scores, but they reduce semantic accuracy.

Another frequent issue arises with text as visual input in the dataset. Multiple annotations relate to on-screen text, but the model, while recognizing the presence of text, cannot transcribe it and often hallucinates words. A similar limitation appears with static textual content: the dataset may include samples where the displayed text does not change, and therefore no events are captured. Consequently, the model receives insufficient signal to generate an informative output.

The last example shows errors requiring broader contextual reasoning. The ground truth mentions a bicycle, which is not explicitly visible in the event stream. Instead, the annotation relies on commonsense knowledge, such as inferring the presence of a bicycle from the helmet. Our model correctly identified the field in the background, but misclassified the subject as male, reflecting a known challenge in vision-language models: social biases. In particular, gender recognition is affected by the underrepresentation of women in training data, leading to systematic misclassifications.

From empirical observation of the generated captions, the outputs of this relatively small model, with fewer than one billion parameters and trained on a limited dataset, appear to be of surprisingly high quality. The model is able to capture both the general topic of the event stream and the associated action, the latter being especially important in the context of event cameras.

After the cross-modal finetuning, we attempted to unfreeze the event encoder and train it jointly with the aligner. This, however, resulted in overfitting, as the event encoder began to diverge significantly from the previously learned representations. Direct comparison with existing event-based VLMs on our event-text dataset was not possible: EventVL has not been made publicly available, and EventGPT, whose inference model was released during the course of this project, does not yet support event streams longer than 100 ms.

Chapter 4

Discussion

Advantages. Our method presents several key benefits over existing approaches. Most importantly, it eliminates the need for frame conversion, a step common to all prior event-based vision-language models that inherently sacrifices event data's native sparsity, compromises fine-grained temporal and polarity information, and introduces sensitivity to the number of events per frame. In contrast, our work explicitly clusters events by their timestamps and polarities into Group Tokens, thereby preserving the raw signal's integrity without incurring substantial computational overhead. Building on this representation, the model achieves data efficiency through its GET encoder and maintains a compact architecture, leaving clear headroom for further scaling. Furthermore, our self-supervised framework is uniquely self-contained: it operates independently of image-domain proxies such as image pre-trained backbones, frame-based event representations, and paired RGB images, unlike the majority of previous event-based pre-training paradigms. On the data side, we adopt an economical construction: we synthesize event-text pairs by simulating events over already labeled datasets, thus avoiding the expense of manual annotation or heavy LLM-driven pipelines used in earlier studies.

Disadvantages. Despite its advantages, our approach is not without limitations. Firstly, the SimCLR-inspired pre-training stage is computationally demanding, requiring long training times with large batch sizes and multi-view augmentation. Additionally, the lightweight aligner module may serve as a capacity bottleneck, potentially limiting the model's overall captioning performance. Concerning the dataset, the YouTube-sourced video clips tend to be generic in content, and each video is paired with only a single caption, reducing robustness to linguistic variation. With respect to event representation, the tokenization process introduces critical hyperparameters for temporal and spatial binning, making the model sensitive to these design choices. Lastly, the generated captions currently yield subpar scores on standard automatic metrics, suggesting substantial room for improvement in event stream captioning.

Applications. The integration of an event-based VLM would enable systems equipped with event cameras to produce human-understandable textual descriptions of the captured dynamic scenes. This is particularly valuable in scenarios requiring low latency and high dynamic range, such as autonomous navigation,

robotic sensing, and high-speed inspection. In these settings, the technology would deliver immediate, actionable insights that support situational awareness and human-in-the-loop decision-making.

4.1 Conclusion

We introduce, to our knowledge, the first event-native vision-language model that entirely eliminates frame conversion and leverages event-based pre-training to ground language generation in event data.

Inspired by state-of-the-art methods from the image domain, we developed a two-stage pre-training framework specifically adapted for event-based vision. Both phases utilized a corpus of 172,000 samples, aggregated from public event-based classification datasets spanning diverse data properties and object categories. The first stage employed unimodal self-supervised contrastive pre-training, enabling the encoder to capture the unique characteristics of event data and learn broad, transferable representations for downstream tasks. The second stage, based on multimodal contrastive pre-training, further refined these representations, as demonstrated by gains in linear probe classification and retrieval metrics. Moreover, it reduced the modality gap between event streams and language by incorporating labels that were unused in the previous stage. Despite its conceptual simplicity, our approach shows superior performance over existing event-based pre-training methods on the CIFAR10-DVS benchmark: it achieves 83.9% accuracy in a linear probe evaluation with a backbone of comparable size, surpassing 78.0% by [74] and 78.6% by [33]. As a result, we obtained an event encoder that operates without reliance on frame representations or pre-existing image-domain knowledge, learns transferable representations from event data, and generalizes effectively to a diverse set of tasks including classification, event-to-event and text-to-event retrieval, as well as event stream captioning.

To facilitate research on event-aware vision-language models, we built an event-text dataset by simulating event streams from the InternVid dataset and adapting its existing annotations. Using this resource, we trained an alignment module connecting a frozen event encoder to a pre-trained text decoder, yielding a captioning model for event streams. While automatic metrics scored modestly, qualitative inspection confirmed the model captures both the overall context and the precise dynamic action in event sequences — the latter being especially important for event camera applications. This remains impressive given the modest scale of both the model and the dataset relative to contemporary large-scale vision-language systems.

4.2 Future Work

This study opens up several promising avenues for exploration. Building on the foundations laid here, future investigations can extend the model's capabilities, refine its evaluation, and broaden its applications across a wider range of event-based vision-language tasks.

Scaling the event encoder. A natural next step is to scale the event encoder, thereby increasing its representational capacity. Larger and deeper encoders may allow the model to capture more complex spatiotemporal structures inherent in event streams, ultimately improving captioning quality.

Enhancing the aligner module. While the lightweight aligner ensures efficiency, it may also constitute a bottleneck. Exploring more expressive architectures, such as Q-Former or other transformer-based alignment mechanisms, could provide richer cross-modal interactions and alleviate performance limitations.

Contributing new benchmarks. The YouTube-sourced clips make our dataset relatively generic in content, and the availability of only a single caption per video limits its linguistic diversity. To enable more robust evaluation, benchmarks with human-annotated captions such as COCO [45] could be incorporated, as they provide multiple captions per sample. A neuromorphic COCO benchmark could then be constructed by generating simple video sequences through translations, rotations, or affine transformations of images, followed by simulating the corresponding events. This approach would produce a high-quality, multi-caption dataset tailored to event-based research. Similarly, neuromorphic variants of established VQA datasets could be developed by converting image-based tasks into event-based form.

Expanding to new tasks. Beyond event stream captioning, tackling more demanding multimodal reasoning tasks such as event-based visual question answering offers a compelling path forward. This extension would require the model to go beyond descriptive generation, engaging instead in reasoning and compositional understanding grounded in event data.

Broader comparisons. Finally, future work should also include systematic comparisons with alternative approaches to event-based text generation, in order to position our method within the broader landscape and better highlight its unique strengths and limitations.

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